JESS Thermodynamic Database v8.9

Arsenic

24-Jan-23

Reaction No. 4716

AsO4-3 + 2<H2O> + 2<e-1> = AsO2-1 + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.71(2SF)	0	CRV	6330

Reaction No. 4717

0.5<As2O3(Claud.,s)> + 3<H+1> + 3<e-1> = As(s) + 1.5<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.9(3SF)	3	EES	
Note (1): Inserted to provide ambient reference point							
2	25:45	0	Inf. Dilution	lgK:11.9(0.05SD)	5	MEF	140
3	25	0.2:1	HClO4	E:0.1(0.05SD)	0	MEF	775
Note (3): Low wgt for inter- and intra-reaction consistency							

Reaction No. 4718

H+1_AsO2-1 + 3<H+1> + 3<e-1> = As(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.248(3SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.24(2SF)	3	CRV	22551

Reaction No. 4719

AsO2-1 + 3<e-1> + 2<H2O> = As(s) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-34.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.68(2SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.68(2SF)	0	CRV	22551

Reaction No. 4720

As(s) + 3<H+1> + 3<e-1> = AsH3(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.1(3SF)	1	ELC	

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2	25	0	Unknown	lgK:-31.(2SF)	0	ENS	775
Note (2): Low wgt for inter-reaction consistency							
3	25	0	Inf. Dilution	E:-0.608(3SF)	0	CRV	6330
Note (3): Low wgt for inter-reaction consistency							
4	25	0	Inf. Dilution	E:-0.225(3SF)	0	CRV	22551

Reaction No. 4739

H+1(3)_AsO4-3 + 2<e-1> + 2<H+1> = H+1_AsO2-1 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.560(3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:0.559(3SF)	0	ENS	7276
4	25	0	Inf. Dilution	E:0.56(2SF)	0	CRV	22551

Reaction No. 9382

As+3_OH-1(2) + H+1 + NTA-3 = As+3_OH-1_NTA-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	24	0.1	Na2SO4	lgK:15.3(0.2SD)	4	MGL	1118
2	25	0.1	Na2SO4	lgK:15.6(0.2SD)	4	MPL	1118

Reaction No. 12538

As+3_OH-1(2) + H+1_NTA-3 = As+3_H+1_OH-1(2)_NTA-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	24	0.1	Na2SO4	lgK:15.33(4SF)	4	MGL	999
2	25	0.1	Na2SO4	lgK:15.58(4SF)	4	MPL	999

Reaction No. 20118

As+3_OH-1(3) + 2<H+1(2)_Tiron-4> = As+3_Tiron-4(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:-8.186(4SF)	6	MGL	999

Reaction No. 20745

As+3_OH-1(3) + H+1_Cat-2 = As+3_OH-1(2)_Cat-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.34(3SF)	6	ENS	773

Reaction No. 20746

As+3_OH-1(3) + H+1(2)_Cat-2 + H+1_Cat-2 = As+3_Cat-2(2) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.83(3SF)	6	ENS	773

Reaction No. 21475

As+3_H+1(-1)_DMannitol_OH-1(2) = As+3_H+1(-2)_DMannitol_OH-1(2) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:8.44(3SF)	6	MOR	999

Reaction No. 22055

H+1 + Arsenazo3-8 = H+1_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:11.85(4SF)	0	MSP	906
2	20	0.2	NaNO3	lgK:12.33(4SF)	0	ENS	1841
3	23			lgK:12.4(0.2SD)	0	MSP	906
4	25	0	Inf. Dilution	lgK:15.7(3SF)	1	ELC	
5	25	0.1	NaClO4	lgK:10.99(4SF)	0	MGL	2878
6	25	1	KCl	lgK:12.6(0.06SD)	1	MSP	1841

Reaction No. 22056

2<H+1> + Arsenazo3-8 = H+1(2)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:23.2(0.04SD)	4	MSP	1841

Reaction No. 22057

3<H+1> + Arsenazo3-8 = H+1(3)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.6(3SF)	1	ELC	
2	25	1	KCl	lgK:32.6(0.04SD)	2	MSP	1841

Reaction No. 22058

4<H+1> + Arsenazo3-8 = H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.4(3SF)	1	ELC	
2	25	1	KCl	lgK:40.1(0.04SD)	3	MSP	1841

Reaction No. 22059

5<H+1> + Arsenazo3-8 = H+1(5)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:45.9(0.05SD)	4	MSP	1841

Reaction No. 22060

6<H+1> + Arsenazo3-8 = H+1(6)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:48.5(0.07SD)	4	MSP	1841

Reaction No. 22061

H+1_Arsenazo3-8 + H+1 = H+1(2)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:9.30(3SF)	4	MSP	906
2	23			lgK:10.2(0.05SD)	4	MSP	906
3	25	0.1	NaClO4	lgK:10.58(4SF)	5	MGL	2878
4	25	1	KCl	lgK:10.7(0.05SD)	3	MSP	1841

Reaction No. 22062

H+1(2)_Arsenazo3-8 + H+1 = H+1(3)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:7.64(3SF)	4	MSP	906
2	23			lgK:8.4(0.05SD)	4	MSP	906
3	25	0.1	NaClO4	lgK:8.79(3SF)	5	MGL	2878
4	25	1	KCl	lgK:9.3(0.04SD)	4	MSP	1841

Reaction No. 22063

H+1(3)_Arsenazo3-8 + H+1 = H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:6.77(3SF)	4	MSP	906
2	20	0.2	NaNO3	lgK:7.48(3SF)	3	ENS	1841
3	23			lgK:7.0(0.03SD)	0	MSP	906
4	25	0.1	NaClO4	lgK:6.70(3SF)	0	MGL	2878
5	25	1	KCl	lgK:7.5(0.04SD)	2	MSP	1841

Reaction No. 22064

H+1(4)_Arsenazo3-8 + H+1 = H+1(5)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	19:21			lgK:5.05(3SF)	0	MSP	906
2	20	0.2	NaNO3	lgK:5.35(3SF)	0	ENS	1841
3	23			lgK:4.96(0.02SD)	0	MSP	906
4	25	0	Inf. Dilution	lgK:8.8(2SF)	1	ELC	
5	25	0.1	NaClO4	lgK:4.29(3SF)	0	MGL	2878
6	25	1	KCl	lgK:5.8(0.05SD)	2	MSP	1841

Reaction No. 22065

H+1(5)_Arsenazo3-8 + H+1 = H+1(6)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:3.64(3SF)	4	MSP	906
2	20	0.2	NaNO3	lgK:2.41(3SF)	6	ENS	1841
3	23			lgK:3.53(0.01SD)	4	MSP	906
4	25	0.1	NaClO4	lgK:2.60(3SF)	5	MGL	2878
5	25	1	KCl	lgK:2.6(0.07SD)	4	MSP	1841

Reaction No. 22066

5<H+1> + Arsenazo3-8 + Cu+2 = Cu+2_H+1(5)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:51.3(0.03SD)	4	MSP	1841

Reaction No. 22067

3<H+1> + Arsenazo3-8 + Cu+2 = Cu+2_H+1(3)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:42.6(0.03SD)	4	MSP	1841

Reaction No. 22068

Arsenazo3-8 + Cu+2 = Cu+2_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:16.2(0.05SD)	4	MSP	1841

Reaction No. 22069

H+1(6)_Arsenazo3-8 + H+1 = H+1(7)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:2.96(3SF)	0	MSP	906
2	20	0.2	NaNO3	lgK:2.41(3SF)	3	ENS	1841
3	23			lgK:2.9(0.2SD)	0	MSP	906

4	25	0	Inf. Dilution	lgK:2.5(2SF)	1	ELC	
5	25	0.1	NaClO4	lgK:1.3(2SF)	0	MGL	2878

Reaction No. 22070							
H+1 + Arsenazo1-6 = H+1_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:12.66(4SF)	0	MSP	906
2	23			lgK:11.8(3SF)	0	MSP	906
3	25	0.1	KNO3	lgK:10.15(4SF)	5	MGL	906
4	25	1	KCl	lgK:12.11(0.02SD)	6	MGL	1683

Reaction No. 22071							
H+1_Arsenazo1-6 + H+1 = H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:10.40(4SF)	0	MSP	906
2	23	0	Inf. Dilution	lgK:9.31(3SF)	2	MPT	906
3	23			lgK:10.0(3SF)	4	MSP	906
4	25	0.1	KNO3	lgK:7.65(3SF)	0	MGL	906
5	25	0.1	KNO3	lgK:9.98(3SF)	3	MGL	906
6	25	1	KCl	lgK:9.8(0.03SD)	3	MSP	1841
7	35	0.1	HNO3	lgR:10.50(4SF)	5	MGL	2332

Reaction No. 22072							
H+1(2)_Arsenazo1-6 + H+1 = H+1(3)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:7.85(3SF)	5	MSP	906
2	23	0	Inf. Dilution	lgK:7.89(3SF)	5	MPT	906
3	23			lgK:8.6(2SF)	4	MSP	906
4	25	0.1	KNO3	lgK:2.93(3SF)	0	MGL	906
5	25	0.1	KNO3	lgK:7.57(3SF)	5	MGL	906
6	25	1	KCl	lgK:7.6(0.03SD)	4	MSP	1841
7	35	0.1	HNO3	lgR:5.35(3SF)	5	MGL	2332

Reaction No. 22073							
H+1(5)_Arsenazo1-6 + H+1 = H+1(6)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:0.68(2SF)	0	MSP	906

2	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EES	
3	25	1	KCl	lgK:2.65(0.02SD)	0	MSP	1841

Reaction No. 22958

As+3_OH-1(4) + DMannitol = As+3_H+1(-2)_DMannitol_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:1.07(3SF)	6	MOR	999

Reaction No. 22959

As+3_H+1_OH-1(4) + DMannitol = As+3_H+1(-1)_DMannitol_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:0.38(2SF)	6	MOR	999

Reaction No. 26713

2<H+1> + Arsenazo1-6 = H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.9(3SF)	1	ELC	
2	25	1	KCl	lgK:21.87(0.02SD)	3	MGL	1683

Reaction No. 26714

3<H+1> + Arsenazo1-6 = H+1(3)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.2(3SF)	1	ELC	
2	25	1	KCl	lgK:29.51(0.02SD)	3	MGL	1683

Reaction No. 26715

4<H+1> + Arsenazo1-6 = H+1(4)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.9(3SF)	1	ELC	
2	25	1	KCl	lgK:32.15(0.006SD)	6	MGL	1683

Reaction No. 26716

5<H+1> + Arsenazo1-6 = H+1(5)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.6(3SF)	1	EES	
2	25	1	KCl	lgK:34.30(0.006SD)	3	MGL	1683

Reaction No. 26717

H+1 + Arsenazo1-6 + Cu+2 = Cu+2_H+1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.0(3SF)	1	ELC	
2	25	1	KCl	lgK:23.5(0.03SD)	6	MGL	1683

Reaction No. 26718

Arsenazo1-6 + Cu+2 = Cu+2_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:16.8(0.06SD)	6	MGL	1683

Reaction No. 26719

2<H+1> + Arsenazo1-6 + 2<Cu+2> = Cu+2(2)_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.5(3SF)	1	ELC	
2	25	1	KCl	lgK:31.9(0.03SD)	2	MGL	1683

Reaction No. 26720

Cu+2 + H+1_Arsenazo1-6 = Cu+2_H+1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:11.4(0.04SD)	6	MGL	1683
2	35	0.1	HNO3	lgK:13.60(4SF)	3	MGL	2332

Reaction No. 26721

2<Cu+2> + H+1(2)_Arsenazo1-6 = Cu+2(2)_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:10.0(0.04SD)	5	ENS	1683

Reaction No. 27214

H+1 + MeArson-2 = H+1_MeArson-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:7.82(3SF)	3	MGL	2399
2	25	1	KCl	lgK:8.53(3SF)	3	MGL	2399

Reaction No. 27215

H+1_MeArson-2 + H+1 = H+1(2)_MeArson-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:4.58(3SF)	3	MGL	2399

Reaction No. 27219							
H+1 + Cacodylic-1 = H+1_Cacodylic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:6.24(3SF)	4	MMX	2852
2	25	0	Inf. Dilution	lgK:6.27(3SF)	5	CRV	6330
3	25	1	KCl	lgK:6.15(3SF)	3	MGL	2399
4	25	0.1	HCl	lgR:6.17(3SF)	5	EDH	1454

Reaction No. 28562							
As+3_OH-1(3) + Galactitol = As+3_H+1(-2)_Galactitol_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-8.30(3SF)	5	CRV	2556

Reaction No. 28568							
As+3_OH-1(3) + DRibose = As+3_H+1(-2)_DRibose_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.50(3SF)	4	CRV	2556

Reaction No. 28575							
As+3_OH-1(3) + DXylose = As+3_H+1(-2)_DXylose_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-8.39(3SF)	4	CRV	2556

Reaction No. 28579							
As+3_OH-1(3) + DArabinose = As+3_H+1(-2)_DArabinose_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-7.85(3SF)	4	CRV	2556

Reaction No. 28583							
As+3_OH-1(3) + LArabinose = As+3_H+1(-2)_LArabinose_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-7.89(3SF)	4	CRV	2556

Reaction No. 28588							
As+3_OH-1(3) + DFructose = As+3_H+1(-2)_DFructose_OH-1(2) + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-8.39(3SF)	4	CRV	2556

Reaction No. 28591

As+3_OH-1(3) + LSorbose = As+3_H+1(-2)_LSorbose_OH-1(2) + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-8.05(3SF)	4	CRV	2556

Reaction No. 28597

As+3_OH-1(3) + DGlucose = As+3_H+1(-2)_DGlucose_OH-1(2) + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-9.21(3SF)	4	CRV	2556

Reaction No. 28598

As+3_OH-1(4) + 2<DGlucose> = As+3_H+1(-4)_DGlucose(2)_OH-1 + H+1 + 3<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-5.12(3SF)	4	CRV	2556

Reaction No. 28601

As+3_OH-1(3) + DMannose = As+3_H+1(-2)_DMannose_OH-1(2) + H+1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-6.91(3SF)	4	CRV	2556

Reaction No. 28602

As+3_OH-1(3) + 2<DMannose> = As+3_H+1(-4)_DMannose(2) + H+1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-6.16(3SF)	4	CRV	2556

Reaction No. 31845

As(s) + Cu(s) + H2(g) + 2<O2(g)> - e-1 = Cu+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-698.17(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-855.13(5SF)kJ	5	CRV	29482

Reaction No. 32692

H+1(7)_Arsenazo3-8 + H+1 = H+1(8)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:1.89(3SF)	0	MSP	906
2	23			lgK:0.8(0.04SD)	0	MSP	906
3	25	0.1	NaClO4	lgK:1.3(2SF)	1	MGL	2878

Reaction No. 33116

La(s) + As(s) + 2<O2(g)> = La+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:253.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:185.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:145.0(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:118.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-347.91(5SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-372.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-371.8(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-371.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-370.7(4SF)	0	MTD	6422

Reaction No. 34301

Pt+2_Cl-1(4) + AsAsAsTriEtBu3enylAs+1 = Pt+2_AsAsAsTriEtBu3enylAs+1_Cl-1(3) + Cl-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	30	2	NaCl	lgR:3.95(3SF)	5	MSP	931
2	44.8	2	NaCl	lgR:3.85(3SF)	5	MSP	931
3	60	2	NaCl	lgR:3.74(3SF)	5	MSP	931
4	30:60	2	NaCl	dH:-3.(0.3SD)	5	MTD	931

Reaction No. 34332

Triphenylarsine + Ag+1 = Ag+1_Triphenylarsine

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:3.56(0.01SD)	5	MIS	2872
2	25	0.1	DMSO NH4ClO4	dH:-34.5(0.08SD)kJ	5	MCL	2872
3	25		Methanol:Water 55.6:44.4Wgt	lgK:5.81(3SF)	5	MSP	931
4	25		Methanol:Water 75.4:24.6Wgt	lgK:5.70(3SF)	5	MSP	931
5	25	0.1	Pyridine Et4NClO4	lgK:1.442(4SF)	5	MIS	4714
6	25	0.1	Pyridine Et4NClO4	dH:-15.6(3SF)kJ	4	MCL	4714

Reaction No. 34333

Hg+2_Cl-1(2) + Triphenylarsine = Hg+2_Triphenylarsine_Cl-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Methanol:Water 75.4:24.6Wgt NaClO4	lgK:1.11(3SF)	5	MSP	931

Reaction No. 34334

Hg+2_Triphenylarsine_Cl-1 + Triphenylarsine = Hg+2_Triphenylarsine(2) + Cl-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Methanol:Water 75.4:24.6Wgt NaClO4	lgK:0.83(2SF)	5	MSP	931

Reaction No. 34348

As+3_H+1(-1)_DDC-1_Dithizone-1 + 2<H+1_DDC-1> = As+3_DDC-1(3) + H+1_Dithizone-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20		CCl4	lgK:7.9(0.06SD)	5	MSP	931

Reaction No. 34716

As+3_OH-1(4) + 2<DFructose> = As+3_H+1(-4)_DFructose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:1.08(3SF)	5	MGL	931

Reaction No. 34717

As+3_OH-1(4) + DFructose = As+3_H+1(-2)_DFructose_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.1	KCl	lgK:0.779(3SF)	5	MGL	931
2	25	0.1	KCl	lgK:0.739(3SF)	5	MGL	931
3	35	0.1	KCl	lgK:0.724(3SF)	5	MGL	931
4	45	0.1	KCl	lgK:0.703(3SF)	5	MGL	931
5	25	0.1	KCl	dH:-1.02(3SF)	5	ENS	931

Reaction No. 34722

H+1(2)_AsO3-3 + DFructose = AsO+1_H+1(-2)_DFructose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:0.77(2SF)	3	MGL	140

Reaction No. 34727

H+1(2)_AsO3-3 + DGalactose = AsO+1_H+1(-2)_DGalactose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.29(2SF)	5	MGL	931

Reaction No. 34734

H+1(2)_AsO3-3 + DGlucose = AsO+1_H+1(-2)_DGlucose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.16(2SF)	5	MGL	140

Reaction No. 34738

As+3_OH-1(4) + DMannose = As+3_H+1(-2)_DMannose_OH-1(2) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:2.22(3SF)	5	MGL	931

Reaction No. 34739

As+3_OH-1(4) + 2<DMannose> = As+3_H+1(-4)_DMannose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:2.97(3SF)	5	MGL	931

Reaction No. 34744

H+1(2)_AsO3-3 + DMannose = AsO+1_H+1(-2)_DMannose + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.36(2SF)	5	MGL	140

Reaction No. 37536

AsO+1_OH-1(2) + H+1(2)_Cat-2 = AsO+1_Cat-2 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:2.04(3SF)	4	MGL	140

Reaction No. 40762

As+3_OH-1(4) + 2<LRhamnose> = As+3_H+1(-4)_LRhamnose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:0.68(2SF)	5	MGL	931

Reaction No. 40765

As+3_OH-1(4) + 2<LSorbose> = As+3_H+1(-4)_LSorbose(2) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:1.08(3SF)	5	MGL	931

Reaction No. 42495							
Al+3_H+1(2)_Arsenazo3-8 + 2<H+1> = Al+3 + H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.173(4SF)	5	MSP	4201

Reaction No. 42496							
Al+3_OH-1 + H+1(5)_Arsenazo3-8 = Al+3_H+1(3)_OH-1_Arsenazo3-8 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:0.468(3SF)	5	MSP	4201

Reaction No. 43695							
H+1 + Ascorbic-2 + As+3_OH-1(2) = As+3_H+1_OH-1(2)_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Na2SO4	lgK:18.8(0.1SD)	5	MPL	906

Reaction No. 44824							
As+3_OH-1(2) + H+1 + EDTA-4 = As+3_H+1_OH-1(2)_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:19.3(0.1SD)	3	MPL	5759
2	24	0.1	Na2SO4	lgK:19.6(0.1SD)	3	MPT	5759
3	25	0	Inf. Dilution	lgK:19.5(3SF)	1	ESC	

Reaction No. 45556							
H+1(3)_Arsenazo1-6 + H+1 = H+1(4)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:2.52(3SF)	4	MSP	906
2	23			lgK:3.4(2SF)	0	MSP	906
3	25	0	Inf. Dilution	lgK:7.3(2SF)	1	ELC	
4	25	0.1	KNO3	lgK:2.97(3SF)	5	MGL	906
5	35	0.1	HNO3	lgR:2.10(3SF)	5	MGL	2332

Reaction No. 45557							
H+1(4)_Arsenazo1-6 + H+1 = H+1(5)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:1.39(3SF)	0	MSP	906
2	23			lgK:2.6(2SF)	0	MSP	906
3	25	0	Inf. Dilution	lgK:0.6(1SF)	1	ELC	

4	35	0.1	HNO3	lgR:2.00(3SF)	2	MGL	2332

Reaction No. 45558

Ba+2 + H+1_Arsenazo1-6 = Ba+2_H+1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.15(3SF)	4	MGL	906

Reaction No. 45559

Arsenazo1-6 + Ba+2 = Ba+2_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.22(3SF)	4	MGL	906

Reaction No. 45560

Ca+2 + H+1_Arsenazo1-6 = Ca+2_H+1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.09(3SF)	4	MGL	906

Reaction No. 45561

Arsenazo1-6 + Ca+2 = Ca+2_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.20(3SF)	4	MGL	906

Reaction No. 45562

Cd+2 + H+1_Arsenazo1-6 = Cd+2_H+1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Acetate buffer	lgK:8.5(0.04SD)	4	MSP	906
2	19:21	0.1	Unknown	lgK:8.4(0.09SD)	4	MSP	906

Reaction No. 45563

Ce+3 + H+1(2)_Arsenazo1-6 = Ce+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.50(0.02SD)	5	MSP	906

Reaction No. 45564

Dy+3 + H+1(2)_Arsenazo1-6 = Dy+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.27(0.01SD)	5	MSP	906

Reaction No. 45565							
Er+3 + H+1(2)_Arsenazo1-6 = Er+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.28(0.01SD)	5	MSP	906

Reaction No. 45566							
Eu+3 + H+1(2)_Arsenazo1-6 = Eu+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.36(0.02SD)	5	MSP	906

Reaction No. 45567							
Gd+3 + H+1(2)_Arsenazo1-6 = Gd+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.3(0.04SD)	1	MSP	906
2	23	0	Inf. Dilution	lgK:8.2(0.08SD)	1	MSP	906

Reaction No. 45568							
Ho+3 + H+1(2)_Arsenazo1-6 = Ho+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.3(0.03SD)	5	MSP	906

Reaction No. 45569							
La+3 + H+1(2)_Arsenazo1-6 = La+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.64(0.02SD)	5	MSP	906

Reaction No. 45570							
Lu+3 + H+1(2)_Arsenazo1-6 = Lu+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.42(0.01SD)	5	MSP	906

Reaction No. 45571							
Nd+3 + H+1(2)_Arsenazo1-6 = Nd+3_H+1(2)_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.7(0.04SD)	5	MSP	906

Reaction No. 45572
Pr+3 + H+1(2)_Arsenazo1-6 = Pr+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.50(0.02SD)	5	MSP	906

Reaction No. 45573

Sm+3 + H+1(2)_Arsenazo1-6 = Sm+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.4(0.03SD)	5	MSP	906

Reaction No. 45574

Tb+3 + H+1(2)_Arsenazo1-6 = Tb+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.28(0.01SD)	5	MSP	906

Reaction No. 45575

Tm+3 + H+1(2)_Arsenazo1-6 = Tm+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.33(0.02SD)	5	MSP	906

Reaction No. 45576

Y+3 + H+1(2)_Arsenazo1-6 = Y+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:9.6(0.03SD)	5	MSP	906

Reaction No. 45577

Yb+3 + H+1(2)_Arsenazo1-6 = Yb+3_H+1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Unknown	lgK:8.4(0.03SD)	5	MSP	906

Reaction No. 45578

H+1(4)_Arsenazo1-6 + Ga+3 = Ga+3_H+1(2)_Arsenazo1-6 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.7(0.2SD)	4	MSP	906

Reaction No. 45579

H+1(4)_Arsenazo1-6 + In+3 = In+3_H+1(2)_Arsenazo1-6 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.(0.3SD)	4	MSP	906

Reaction No. 45580							
Mg+2 + H+1_Arsenazo1-6 = Mg+2_H+1_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.58(3SF)	4	MGL	906

Reaction No. 45581							
Arsenazo1-6 + Mg+2 = Mg+2_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:5.53(3SF)	4	MGL	906

Reaction No. 45582							
Sr+2 + H+1_Arsenazo1-6 = Sr+2_H+1_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.41(3SF)	4	MGL	906

Reaction No. 45583							
Arsenazo1-6 + Sr+2 = Sr+2_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KNO3	lgK:4.39(3SF)	4	MGL	906

Reaction No. 45584							
Zn+2 + H+1_Arsenazo1-6 = Zn+2_H+1_Arsenazo1-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21	0.1	Acetate buffer	lgK:9.9(0.03SD)	4	MSP	906
2	19:21	0.1	Unknown	lgK:9.9(0.07SD)	4	MSP	906

Reaction No. 45640							
Ce+3 + H+1(4)_Arsenazo3-8 = Ce+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:14.9(0.1SD)	4	MSP	906

Reaction No. 45641							
Dy+3 + H+1(4)_Arsenazo3-8 = Dy+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.7(0.03SD)	4	MSP	906

Reaction No. 45642
Er+3 + H+1(4)_Arsenazo3-8 = Er+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.4(0.1SD)	4	MSP	906

Reaction No. 45643							
Eu+3 + H+1(4)_Arsenazo3-8 = Eu+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.(0.3SD)	4	MSP	906

Reaction No. 45644							
Gd+3 + H+1(4)_Arsenazo3-8 = Gd+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.7(0.1SD)	4	MSP	906

Reaction No. 45645							
Ho+3 + H+1(4)_Arsenazo3-8 = Ho+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.6(0.09SD)	4	MSP	906

Reaction No. 45646							
La+3 + H+1(4)_Arsenazo3-8 = La+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:14.9(0.07SD)	4	MSP	906

Reaction No. 45647							
Lu+3 + H+1(4)_Arsenazo3-8 = Lu+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.1(0.1SD)	4	MSP	906

Reaction No. 45648							
Nd+3 + H+1(4)_Arsenazo3-8 = Nd+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.(0.3SD)	4	MSP	906

Reaction No. 45649							
Pr+3 + H+1(4)_Arsenazo3-8 = Pr+3_H+1(4)_Arsenazo3-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.4(0.1SD)	4	MSP	906

Reaction No. 45650

Sm+3 + H+1(4)_Arsenazo3-8 = Sm+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.(0.3SD)	4	MSP	906

Reaction No. 45651

Tb+3 + H+1(4)_Arsenazo3-8 = Tb+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.(0.3SD)	4	MSP	906

Reaction No. 45652

Tm+3 + H+1(4)_Arsenazo3-8 = Tm+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.2(0.09SD)	4	MSP	906

Reaction No. 45653

Y+3 + H+1(4)_Arsenazo3-8 = Y+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:16.1(0.1SD)	4	MSP	906

Reaction No. 45654

Yb+3 + H+1(4)_Arsenazo3-8 = Yb+3_H+1(4)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19:21			lgK:15.0(0.2SD)	4	MSP	906

Reaction No. 45655

Gd+3 + H+1(5)_Arsenazo3-8 = Gd+3_H+1(5)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:5.4(2SF)	4	MSP	906

Reaction No. 45656

UO2+2 + H+1(6)_Arsenazo3-8 = UO2+2_H+1(6)_Arsenazo3-8

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.6	HCl	lgK:4.22(3SF)	4	MPL	906

Reaction No. 45657

UO2+2 + 2<H+1(6)_Arsenazo3-8> = UO2+2_H+1(12)_Arsenazo3-8(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	23	0.6	HCl	lgK:8.11(3SF)	4	MPL	906

Reaction No. 45659

2<Y+3> + 2<H+1(4)_Arsenazo3-8> = Y+3(2)_H+1(8)_Arsenazo3-8(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.3:12	KCl	lgK:22.56(4SF)	4	MDS	906

Reaction No. 45994

As+3_OH-1(2) + Citric-3 = As+3_OH-1(2)_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:9.3(2SF)	4	CRV	2556

Reaction No. 45995

As+3_OH-1(2) + H+1_Citric-3 = As+3_H+1_OH-1(2)_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:8.5(2SF)	4	CRV	2556

Reaction No. 53496

AsO4-3 + H+1 = H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:11.85(4SF)	3	CRV	28992
2	10	0	Inf. Dilution	lgK:11.33(4SF)	0	MGL	5469
3	18			lgK:11.53(4SF)	0	MGL	140
4	20			lgK:11.4(3SF)	0	MCO	140
5	20			lgK:11.5(3SF)	0	MSL	140
6	25	0.1:0.3	Unknown	lgK:11.53(4SF)	0	MCL	931
7	25	0	Inf. Dilution	lgK:11.57(4SF)	1	EOR	
8	25	0	Inf. Dilution	lgK:11.50(0.05SD)	4	MGL	5469
9	25	0	Inf. Dilution	lgK:11.29(4SF)	2	CRV	6330
10	25	0	Inf. Dilution	lgK:11.50(4SF)	4	CRV	6478
11	25	0	Inf. Dilution	lgK:12.5(3SF)	0	ENS	7276
12	25	0	Inf. Dilution	lgK:11.52(4SF)	4	CRV	22551
13	25	0	Inf. Dilution	lgK:11.67(0.01SD)	3	EOD	28961
14	25	0	Inf. Dilution	lgK:11.52(4SF)	3	CRV	28992
15	25	0	Inf. Dilution	lgK:11.68(0.02SD)	3	EDH	29940
16	25	0	Inf. Dilution	lgK:11.60(4SF)	4	EOD	31763
17	25	0.1	NaCl	lgK:11.08(0.02SD)	2	MGL	29940

18	25	0.1	Unknown	lgK:11.18(4SF)	2	CRV	552
19	25	0.25	NaCl	lgK:10.79(0.02SD)	5	MGL	29940
20	25	0.5	NaCl	lgK:10.62(0.01SD)	5	MGL	29940
21	25	1	NaCl	lgK:10.51(0.005SD)	5	MGL	29940
22	25			lgK:9.21(3SF)	0	MCN	140
23	35	0	Inf. Dilution	lgK:11.64(0.05SD)	2	MGL	5469
24	50	0	Inf. Dilution	lgK:11.5(3SF)	3	EES	5398
25	50	0	Inf. Dilution	lgK:11.92(4SF)	0	MGL	5469
26	50	0	Inf. Dilution	lgK:11.30(4SF)	3	CRV	28992
27	60	0	Inf. Dilution	lgK:11.4(3SF)	3	EES	5398
28	100	0	Inf. Dilution	lgK:11.4(3SF)	1	ENS	5398
29	100	0	Inf. Dilution	lgK:11.15(4SF)	3	CRV	28992
30	150	0	Inf. Dilution	lgK:11.7(3SF)	0	ENS	5398
31	150	0	Inf. Dilution	lgK:11.25(4SF)	3	CRV	28992
32	200	0	Inf. Dilution	lgK:12.3(3SF)	0	ENS	5398
33	200	0	Inf. Dilution	lgK:11.50(4SF)	3	CRV	28992
34	250	0	Inf. Dilution	lgK:11.9(3SF)	3	CRV	28992
35	10:50	0	Inf. Dilution	dH:5.4(2SF)	0	MGL	5469
36	25	0.1:0.3	Unknown	dH:-4.35(3SF)	2	MCL	931
37	25	0	Inf. Dilution	dH:-4.35(3SF)	4	CRV	6478
38	25	0	Inf. Dilution	dH:-18.3(3SF)kJ	4	EOD	28961
39	25	0.1	NaCl	dH:-18.1(0.04SD)kJ	4	MCL	29940
40	25	0.7	NaCl	dH:-20.3(0.04SD)kJ	4	MCL	29940

Reaction No. 53497

H+1_AsO4-3 + H+1 = H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:7.1(0.05SD)	3	ENS	12
2	0	0	Inf. Dilution	lgK:7.07(3SF)	3	CRV	28992
3	4.8	1	NaClO4	lgK:6.10(3SF)	2	MEF	5398
4	10	0	Inf. Dilution	lgK:7.05(3SF)	3	MGL	5469
5	18			lgK:6.77(3SF)	0	MGL	140
6	20			lgK:5.9(2SF)	0	MSL	140
7	22	1	NaClO4	lgK:5.81(0.05SD)	0	MEF	5037
8	22	1	NaClO4	lgK:5.75(3SF)	0	MEF	5037

Note (8): 0.1M / 0.2M H2AsO4- added

9	25	0.1:0.3	Unknown	lgK:6.77(3SF)	2	MCL	931
10	25	0	Inf. Dilution	lgK:7.08(3SF)	4	MKN	140
11	25	0	Inf. Dilution	lgK:6.98(3SF)	4	MQH	140
12	25	0	Inf. Dilution	lgK:6.94(3SF)	4	MGL	5469
13	25	0	Inf. Dilution	lgK:6.76(3SF)	2	CRV	6330
14	25	0	Inf. Dilution	lgK:6.94(3SF)	4	CRV	6478
15	25	0	Inf. Dilution	lgK:7.2(2SF)	0	ENS	7276
16	25	0	Inf. Dilution	lgK:6.77(3SF)	4	CRV	22551
17	25	0	Inf. Dilution	lgK:6.98(3SF)	4	EOD	28961
18	25	0	Inf. Dilution	lgK:6.97(3SF)	3	CRV	28992
19	25	0	Inf. Dilution	lgK:7.07(3SF)	3	EDH	29940
Note (19): calcualted from overall values							
20	25	0.1	NaCl	lgK:6.62(3SF)	5	MGL	29940
Note (20): calculated from overall values							
21	25	0.1	Unknown	lgK:6.65(3SF)	4	CRV	552
22	25	0.25	NaCl	lgK:6.50(3SF)	5	MGL	29940
Note (22): calculated from overall values							
23	25	0.5	NaCl	lgK:6.38(3SF)	5	MGL	29940
Note (23): calculated from overall values							
24	25	1	Na2SO4	lgK:6.39(3SF)	0	MGL	5036
25	25	1	NaCl	lgK:6.27(3SF)	5	MGL	29940
Note (25): calculated from overall values							
26	25			lgK:4.39(3SF)	0	MCN	140
27	25			lgK:6.80(3SF)	0	MKN	140
28	35	0	Inf. Dilution	lgK:6.87(3SF)	4	MGL	5469
29	50	0	Inf. Dilution	lgK:6.7(2SF)	2	ENS	5398
30	50	0	Inf. Dilution	lgK:6.80(3SF)	3	MGL	5469
31	50	0	Inf. Dilution	lgK:6.96(3SF)	3	CRV	28992
32	60	0	Inf. Dilution	lgK:6.7(2SF)	2	ENS	5398
33	100	0	Inf. Dilution	lgK:6.8(2SF)	0	ENS	5398
34	100	0	Inf. Dilution	lgK:7.16(3SF)	3	CRV	28992
35	150	0	Inf. Dilution	lgK:7.1(2SF)	0	ENS	
36	150	0	Inf. Dilution	lgK:7.54(3SF)	3	CRV	28992
37	200	0	Inf. Dilution	lgK:7.8(2SF)	3	EES	5398
38	200	0	Inf. Dilution	lgK:8.06(3SF)	3	CRV	28992
39	250	0	Inf. Dilution	lgK:8.67(3SF)	3	CRV	28992

40	10:50	0	Inf. Dilution	dH:-2.8(0.1SD)	0	MGL	5469
41	25	0.1:0.3	Unknown	dH:-0.80(2SF)	5	MCL	5035
42	25	0	Inf. Dilution	dH:-0.77(2SF)	4	CRV	6478
43	25	0	Inf. Dilution	dH:-2.86(3SF)kJ	4	EOD	28961
44	25	0.1	NaCl	dH:-6.3(0.45SD)kJ	4	MCL	29940
45	25	0.7	NaCl	dH:-2.9(0.4SD)kJ	4	MCL	29940

Reaction No. 53498							
H+1(2)_AsO4-3 + H+1 = H+1(3)_AsO4-3							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	0	0	Inf. Dilution	lgK:2.20(3SF)	3	CRV	28992
2	10	0	Inf. Dilution	lgK:2.49(0.05SD)	0	MGL	5469
3	18			lgK:2.25(3SF)	0	MGL	140
4	20			lgK:2.7(2SF)	0	MSL	140
5	25	0.1:0.3	Unknown	lgK:2.25(3SF)	2	MCL	931
6	25	0	Inf. Dilution	lgK:2.36(3SF)	2	MCN	140
7	25	0	Inf. Dilution	lgK:2.19(3SF)	4	MGL	140
8	25	0	Inf. Dilution	lgK:2.22(3SF)	4	MQH	140
9	25	0	Inf. Dilution	lgK:2.30(3SF)	4	MGL	288
10	25	0	Inf. Dilution	lgK:2.24(0.01SD)	4	ENS	766
11	25	0	Inf. Dilution	lgK:2.19(0.08SD)	4	MGL	5469
12	25	0	Inf. Dilution	lgK:2.26(3SF)	4	CRV	6330
13	25	0	Inf. Dilution	lgK:2.24(3SF)	4	CRV	6478
14	25	0	Inf. Dilution	lgK:3.5(2SF)	0	ENS	7276
15	25	0	Inf. Dilution	lgK:2.32(3SF)	3	CRV	22551
16	25	0	Inf. Dilution	lgK:2.26(3SF)	3	CRV	28992
17	25	0	Inf. Dilution	lgK:2.42(3SF)	3	EDH	29940
Note (17): calculated from overall values							
18	25	0	Inf. Dilution/m	lgK:2.26(3SF)	2	CRV	24953
19	25	0.01	Unknown/m	lgK:2.17(3SF)	2	CRV	24953
20	25	0.1	NaCl	lgK:2.13(3SF)	5	MGL	29940
Note (20): calculated from overall values							
21	25	0.1	Unknown	lgK:2.15(3SF)	3	CRV	552
22	25	0.1	Unknown/m	lgK:2.04(3SF)	2	CRV	24953
23	25	0.2	Unknown/m	lgK:1.99(3SF)	2	CRV	24953
24	25	0.25	NaCl	lgK:2.27(3SF)	5	MGL	29940

Note (24): calculated from overall values							
25	25	0.5	NaCl	lgK:2.08(3SF)	5	MGL	29940
Note (25): calculated from overall values							
26	25	0.5	Unknown/m	lgK:1.93(3SF)	2	CRV	24953
27	25	1	NaCl	lgK:2.03(3SF)	5	MGL	29940
Note (27): calculated from overall values							
28	25	1	Unknown/m	lgK:1.90(3SF)	2	CRV	24953
29	25	2	Unknown/m	lgK:1.91(3SF)	2	CRV	24953
30	25	3	NaClO4	lgK:2.3(0.05SD)	5	MGL	5038
31	25	3	Unknown/m	lgK:1.96(3SF)	2	CRV	24953
32	25			lgK:2.30(3SF)	0	MCN	140
33	25			lgK:2.32(3SF)	0	MCN	140
34	25			lgK:2.301(4SF)	0	MGL	931
35	35	0	Inf. Dilution	lgK:1.95(0.05SD)	0	MGL	5469
Note (35): Low wgt for internal consistency							
36	50	0	Inf. Dilution	lgK:2.3(2SF)	3	EES	5398
37	50	0	Inf. Dilution	lgK:2.15(0.05SD)	0	MGL	5469
38	50	0	Inf. Dilution	lgK:2.40(3SF)	3	CRV	28992
39	50	0	Inf. Dilution/m	lgK:2.40(3SF)	2	CRV	24953
40	60	0	Inf. Dilution	lgK:2.4(2SF)	3	EES	5398
41	75	0	Inf. Dilution/m	lgK:2.60(3SF)	2	CRV	24953
42	100	0	Inf. Dilution	lgK:2.6(2SF)	0	ENS	5398
43	100	0	Inf. Dilution	lgK:2.85(3SF)	3	CRV	28992
44	100	0	Inf. Dilution/m	lgK:2.84(3SF)	2	CRV	24953
45	150	0	Inf. Dilution	lgK:3.0(2SF)	0	ENS	5398
46	150	0	Inf. Dilution	lgK:3.50(3SF)	3	CRV	28992
47	150	0	Inf. Dilution/m	lgK:3.39(3SF)	1	CRV	24953
48	200	0	Inf. Dilution	lgK:3.5(2SF)	0	ENS	5398
49	200	0	Inf. Dilution	lgK:4.25(3SF)	3	CRV	28992
50	200	0	Inf. Dilution/m	lgK:4.00(3SF)	0	CRV	24953
51	250	0	Inf. Dilution	lgK:5.0(2SF)	3	CRV	28992
52	250	0	Inf. Dilution/m	lgK:4.63(3SF)	0	CRV	24953
53	300	0	Inf. Dilution/m	lgK:5.27(3SF)	0	CRV	24953
54	10:50	0	Inf. Dilution	dH:-0.6(0.1SD)	0	MGL	5469
Note (54): Low wgt for internal consistency							
55	25	0.1:0.3	Unknown	dH:1.69(3SF)	0	MCL	931

56	25	0	Inf. Dilution	dH:-0.60(2SF)	0	MCL	140
Note (56): Low wgt for internal consistency							
57	25	0	Inf. Dilution	dH:1.69(0.01SD)	4	ENS	766
58	25	0	Inf. Dilution	dH:1.69(3SF)	4	CRV	6478
59	25	0.1	NaCl	dH:-5.(0.5SD)kJ	4	MCL	29940
60	25	0.7	NaCl	dH:2.(1.SD)kJ	4	MCL	29940
61	25		Water:D2O .494:.506Wgt	lgK:2.421(4SF)	4	MGL	931
62	25		Water:D2O 1:99Wgt	lgK:2.596(4SF)	4	MGL	931

Reaction No. 54614

Ce+4_OH-1 + H+1(3)_AsO3-3 = Ce+4_H+1(2)_AsO3-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	2.4	1	HClO4	lgK:2.0(0.05SD)	5	MSP	5034
2	5.1	1	HClO4	lgK:2.1(0.05SD)	5	MSP	5034
3	8.4	1	HClO4	lgK:2.0(0.05SD)	5	MSP	5034
4	11.1	1	HClO4	lgK:2.1(0.05SD)	5	MSP	5034
5	25	1	Inf. Dilution	lgK:2.2(0.05SD)	4	MSP	5034
6	2.4:11.1	1	HClO4	dH:2.2(0.1SD)	5	MSP	5034

Reaction No. 54615

Ce+4_OH-1 + H+1(3)_AsO3-3 + H+1 = Ce+4_H+1(3)_AsO3-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Inf. Dilution	lgK:1.7(0.05SD)	4	MSP	5034
2	2.4	1	HClO4	lgR:2.6(0.05SD)	5	MSP	5034
3	5.1	1	HClO4	lgR:2.4(0.05SD)	5	MSP	5034
4	8.4	1	HClO4	lgR:2.2(0.05SD)	5	MSP	5034
5	11.1	1	HClO4	lgR:2.2(0.05SD)	5	MSP	5034
6	2.4:11.1	1	HClO4	dH:-14.2(0.1SD)	5	MSP	5034

Reaction No. 54616

0.5<As2O3(Claud.,s)> + 1.5<H2O> = H+1(3)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.6(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.7(0.05SD)	0	MSL	5032

Reaction No. 54619

2<H+1(3)_AsO3-3> = H+1(6)_AsO3-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0	Inf. Dilution	lgK:-0.9(0.05SD)	4	MSP	5031
2	22	0.5	NaCl	lgK:-0.2(0.05SD)	5	MSP	5031
3	22	1	NaCl	lgK:0.2(0.05SD)	5	MSP	5031
4	25	0	Inf. Dilution	lgK:-0.92(2SF)	3	CRV	552
5	25	0.5	Unknown	lgK:-0.21(2SF)	3	CRV	552
6	25	1	Unknown	lgK:0.21(2SF)	3	CRV	552

Reaction No. 54685							
Zn+2_H+1_AsO4-3_H2O_(s) + 2<H+1> = Zn+2 + H+1(3)_AsO4-3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.5(0.1SD)	4	ENS	5522

Reaction No. 54707							
H+1(2)_AsO3-3 + H+1 = H+1(3)_AsO3-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaCl	lgK:9.2(0.05SD)	0	MGL	5031
2	25	0	Inf. Dilution	lgK:9.29(3SF)	3	CRV	552
3	25	0	Inf. Dilution	lgK:9.23(0.01SD)	4	ENS	766
4	25	0	Inf. Dilution	lgK:9.3(0.05SD)	4	ENS	5018
5	25	0	Inf. Dilution	lgK:9.29(3SF)	4	CRV	6330
6	25	0	Inf. Dilution	lgK:9.22(3SF)	4	ENS	7276
7	25	0	Inf. Dilution	lgK:9.23(3SF)	3	CRV	28992
8	25	0	Inf. Dilution/m	lgK:9.23(3SF)	3	CRV	24953
9	25	0.1	Unknown	lgK:9.14(3SF)	4	CRV	552
10	25	0.1	Unknown/m	lgK:9.13(3SF)	2	CRV	24953
11	25	0.5	Unknown	lgK:9.10(3SF)	4	CRV	552
12	25	0.5	Unknown/m	lgK:9.09(3SF)	2	CRV	24953
13	25	0.59	KCl	lgK:9.1(0.05SD)	3	MGL	5033
14	25	1	Unknown	lgK:9.05(3SF)	4	CRV	552
15	25	1.45	KCl	lgK:9.1(0.05SD)	5	MGL	5033
16	50	0	Inf. Dilution	lgK:8.9(0.05SD)	3	ENS	5
17	50	0	Inf. Dilution	lgK:8.91(3SF)	3	CRV	28992
18	50	0	Inf. Dilution/m	lgK:8.89(3SF)	2	CRV	24953
19	60	0	Inf. Dilution	lgK:8.8(0.05SD)	3	ENS	5

20	75	0	Inf. Dilution/m	lgK:8.69(3SF)	2	CRV	24953
21	100	0	Inf. Dilution	lgK:8.4(0.05SD)	0	ENS	5
22	100	0	Inf. Dilution	lgK:8.60(3SF)	3	CRV	28992
23	100	0	Inf. Dilution/m	lgK:8.57(3SF)	2	CRV	24953
24	150	0	Inf. Dilution	lgK:8.65(3SF)	3	CRV	28992
25	150	0	Inf. Dilution/m	lgK:8.53(3SF)	1	CRV	24953
26	200	0	Inf. Dilution	lgK:8.90(3SF)	3	CRV	28992
27	200	0	Inf. Dilution/m	lgK:8.67(3SF)	1	CRV	24953
28	250	0	Inf. Dilution	lgK:9.3(2SF)	3	CRV	28992
29	250	0	Inf. Dilution/m	lgK:8.91(3SF)	0	CRV	24953
30	25	0.1:0.3	Unknown	dH:-6.6(0.1SD)	5	MCL	5035
31	25	0	Inf. Dilution	dH:-6.56(0.01SD)	4	ENS	766
32	25	0.1	Unknown	dH:-6.6(2SF)	3	CRV	552

Reaction No. 54709

Ag+1(3)_AsO3-3_(s) = 3<Ag+1> + AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-31.3(3SF)	3	CRV	552
2	20	0.1	Unknown	lgK:-31.27(0.24SD)	3	MIS	4593
3	25	0	Inf. Dilution	lgK:-32.5(3SF)	2	EAE	
4	25	0	Inf. Dilution	lgK:-18.35(4SF)	0	CRV	22551

Reaction No. 54710

H+1_AsO3-3 + 5<H+1> + 3<e-1> = As(s) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:12.55(4SF)	0	ENS	140

Reaction No. 54711

AsO4-3 + H2O + 2<e-1> = AsO3-3 + 2<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-22.7(3SF)	0	ENS	140

Reaction No. 54712

Ag+1(3)_AsO4-3_(s) = 3<Ag+1> + AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-22.19(0.19SD)	1	MIS	4593

2	25	0	Inf. Dilution	lgK:-20.21(4SF)	1	CRV	3006
3	25	0	Inf. Dilution	lgK:-21.99(4SF)	3	CRV	6330
Note (3): p. 8-58							
4	25	0	Inf. Dilution	lgK:-19.(2SF)	0	CRV	22551

Reaction No. 54713							
H+1(3)_AsO3-3 + H+1(2)_AsO3-3 = H+1(5)_AsO3-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.5	NaCl	lgK:0.6(0.05SD)	0	MGL	5031
2	22	1	NaCl	lgK:0.8(0.05SD)	0	MGL	5031
3	25	0	Inf. Dilution	lgK:-0.1(1SF)	1	ELC	

Reaction No. 54714							
Al+3_AsO4-3_(s) = Al+3 + AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-15.8(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-34.6(3SF)	1	ELC	

Reaction No. 54715							
Ba+2(3)_AsO4-3(2)_(s) = 3<Ba+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-20.13(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-21.66(4SF)	1	ELC	

Reaction No. 54716							
Ca+2(3)_AsO4-3(2)_(s) = 3<Ca+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-18.2(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-19.8(3SF)	1	ELC	

Reaction No. 54717							
Cd+2(3)_AsO4-3(2)_(s) = 3<Cd+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-32.7(0.05SD)	4	ENS	5468
2	25	0	Inf. Dilution	lgK:-32.13(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:-32.7(3SF)	2	EAE	
4	25	0	Inf. Dilution	lgK:-32.66(4SF)	3	CRV	6330

Note (4): p. 8-58

Reaction No. 54718							
Co+2(3)_AsO4-3(2)_(s) = 3<Co+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-28.1(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-37.4(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-28.1(3SF)	0	CRV	6330
Note (3): p. 8-58							

Reaction No. 54719							
Cu+2(3)_AsO4-3(2)_(s) = 3<Cu+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-35.1(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-38.1(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-35.1(3SF)	0	CRV	
Note (3): http://www.saltlakemetals.com/SolubilityProducts.htm (15/7/14)							
4	25	0	Inf. Dilution	lgK:-35.10(4SF)	0	CRV	6330
Note (4): p. 8-58							

Reaction No. 54720							
Mg+2(3)_AsO4-3(2)_(s) = 3<Mg+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-19.7(0.05SD)	5	ENS	5468
2	25	0	Inf. Dilution	lgK:-19.7(3SF)	5	EES	
Note (2): For a LC boost (only)							
3	25	0	Inf. Dilution	lgK:-19.7(4SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:-19.68(4SF)	3	CRV	
Note (4): http://www.saltlakemetals.com/SolubilityProducts.htm (15/7/14)							

Reaction No. 54721							
Mn+2(3)_AsO4-3(2)_(s) = 3<Mn+2> + 2<AsO4-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-28.7(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-33.0(3SF)	1	ELC	

Reaction No. 54722

Ni+2(3)_AsO4-3(2)_(s) = 3<Ni+2> + 2<AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-25.5(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-39.6(3SF)	1	ELC	

Reaction No. 54723

Pb+2(3)_AsO4-3(2)_(s) = 3<Pb+2> + 2<AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-35.4(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-32.5(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-35.4(3SF)	0	CRV	
Note (3): http://www.saltlakemetals.com/SolubilityProducts.htm (15/7/14)							
4	25	0	Inf. Dilution	lgK:-32.44(4SF)	1	EOR	

Reaction No. 54724

Sr+2(3)_AsO4-3(2)_(s) = 3<Sr+2> + 2<AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-18.1(0.05SD)	4	ENS	5468
2	25	0	Inf. Dilution	lgK:-20.49(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:-18.1(3SF)	2	EAE	
4	25	0	Inf. Dilution	lgK:-18.37(4SF)	3	CRV	6330
Note (4): p. 8-58							

Reaction No. 54725

Zn+2(3)_AsO4-3(2)_(s) = 3<Zn+2> + 2<AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-27.4(0.05SD)	4	ENS	5468
2	25	0	Inf. Dilution	lgK:-28.0(3SF)	1	EES	
3	25	0	Inf. Dilution	lgK:-27.5(3SF)	3	CRV	6330
Note (3): p. 8-58							

Reaction No. 54726

Cr+3_AsO4-3_(s) = Cr+3 + AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-20.1(0.05SD)	4	ENS	5468
2	25	0	Inf. Dilution	lgK:-18.51(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:-20.1(3SF)	2	EAE	

Reaction No. 54727

Fe+3_AsO4-3_(s) = Fe+3 + AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-20.2(0.05SD)	0	ENS	5468
2	25	0	Inf. Dilution	lgK:-19.3(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-20.2(3SF)	0	CRV	
Note (3): http://www.saltlakemetals.com/SolubilityProducts.htm (15/7/14)							
4	25	0	Inf. Dilution	lgK:-19.17(4SF)	1	EOR	

Reaction No. 54728

K+1 + AsF6-1 = K+1_AsF6-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.3(0.05SD)	0	MCN	5039

Reaction No. 54729

H+1 + AsO3CH3-2 = H+1_AsO3CH3-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4(0.05SD)w	4	MPH	5470
2	25	0.025	NaNO3	lgK:8.4(0.05SD)w	5	MPH	5470
3	25	0.05	NaNO3	lgK:8.3(0.05SD)w	5	MPH	5470
4	25	0.075	NaNO3	lgK:8.3(0.05SD)w	5	MPH	5470
5	25	0.1	NaNO3	lgK:8.2(0.05SD)w	5	MPH	5470

Reaction No. 54730

H+1_AsO3CH3-2 + AsO3CH3-2 = H+1_AsO3CH3-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.5(0.05SD)w	4	MPH	5470
2	25	0.025	NaNO3	lgK:4.4(0.05SD)w	5	MPH	5470
3	25	0.05	NaNO3	lgK:4.4(0.05SD)w	5	MPH	5470
4	25	0.075	NaNO3	lgK:4.3(0.05SD)w	5	MPH	5470
5	25	0.1	NaNO3	lgK:4.3(0.05SD)w	5	MPH	5470

Reaction No. 54967

2<As(s)> + 2.5<O2(g)> = As2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:137.0(4SF)	2	ENS	6422

2	27	0	Inf. Dilution	lgK:136.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:95.76(4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:71.63(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:55.58(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-186.9(0.05SD)	3	ENS	5020
7	25	0	Inf. Dilution	dG:-782.3(4SF)kJ	2	CRV	6330
8	25	0	Inf. Dilution	dG:-186.9(4SF)	3	EFE	6422
9	25	0	Inf. Dilution	dG:-781.4(4SF)kJ	4	CRV	12588
10	25	0	Inf. Dilution	dG:-782.45(8.016SD)kJ	2	CRV	14923
11	25	0	Inf. Dilution	dG:-774.76(5SF)kJ	3	CRV	29482
12	25	0	Inf. Dilution	dH:-924.9(4SF)kJ	3	CRV	6330
13	25	0	Inf. Dilution	dH:-221.0(4SF)	3	MTD	6422
14	25	0	Inf. Dilution	dH:-924.87(8.000SD)kJ	4	CRV	14923
15	25	0	Inf. Dilution	dH:-917.59(5SF)kJ	3	CRV	29482
16	127	0	Inf. Dilution	dH:-221.0(4SF)	1	MTD	6422
17	227	0	Inf. Dilution	dH:-221.0(4SF)	0	MTD	6422
18	327	0	Inf. Dilution	dH:-219.9(4SF)	0	MTD	6422

Reaction No. 54968

As(s) + 1.5<H2(g)> + 1.5<O2(g)> = H+1(3)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-154.4(0.05SD)	0	ENS	5020
2	25	0	Inf. Dilution	dG:-152.94(5SF)	4	CRV	12011
3	25	0	Inf. Dilution	dG:-639.9(4SF)kJ	5	CRV	12588
4	25	0	Inf. Dilution	dG:-639.68(4.015SD)kJ	6	CRV	14923
5	25	0	Inf. Dilution	dG:-639.80(5SF)kJ	5	CRV	29482
6	25	0	GAMSPATH	dG:-639.8(4SF)kJ	3	CRV	12638
7	25	0	Inf. Dilution	dH:-177.4(4SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-742.20(4.000SD)kJ	6	CRV	14923
9	25	0	Inf. Dilution	dH:-742.36(5SF)kJ	5	CRV	29482
10	25	0	GAMSPATH	dH:-742.240(6SF)kJ	3	CRV	12638

Reaction No. 54969

As(s) + H2(g) + 1.5<O2(g)> + e-1 = H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-141.8(0.05SD)	0	ENS	5020

2	25	0	Inf. Dilution	dG:-140.33(5SF)	4	CRV	6488
3	25	0	Inf. Dilution	dG:-140.33(5SF)	4	CRV	10174
4	25	0	Inf. Dilution	dG:-587.2(4SF)kJ	5	CRV	12588
5	25	0	Inf. Dilution	dG:-587.08(4.008SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	dG:-587.66(5SF)kJ	5	CRV	29482
7	25	0	GAMSPATH	dG:-587.2(4SF)kJ	3	CRV	12638
8	50	0	Inf. Dilution	dG:-140.99(5SF)	0	CRV	10174
9	75	0	Inf. Dilution	dG:-141.65(5SF)	0	CRV	10174
10	100	0	Inf. Dilution	dG:-142.30(5SF)	0	CRV	10174
11	125	0	Inf. Dilution	dG:-142.95(5SF)	0	CRV	10174
12	150	0	Inf. Dilution	dG:-143.58(5SF)	0	CRV	10174
13	175	0	Inf. Dilution	dG:-144.20(5SF)	0	CRV	10174
14	200	0	Inf. Dilution	dG:-144.79(5SF)	0	CRV	10174
15	225	0	Inf. Dilution	dG:-145.34(5SF)	0	CRV	10174
16	250	0	Inf. Dilution	dG:-145.84(5SF)	0	CRV	10174
17	300	0	Inf. Dilution	dG:-146.51(5SF)	0	CRV	10174
18	25	0	Inf. Dilution	dH:-170.84(5SF)	4	CRV	6488
19	25	0	Inf. Dilution	dH:-714.79(4.000SD)kJ	6	CRV	14923
20	25	0	Inf. Dilution	dH:-714.74(5SF)kJ	5	CRV	29482
21	25	0	GAMSPATH	dH:-714.800(6SF)kJ	3	CRV	12638

Reaction No. 54970

As(s) + 0.5<H2(g)> + 1.5<O2(g)> + 2<e-1> = H+1_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:90.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-125.3(0.05SD)	0	ENS	5020
3	25	0	Inf. Dilution	dG:-507.40(5SF)kJ	0	CRV	29482
4	25	0	GAMSPATH	dG:-518.0(4SF)kJ	0	CRV	12638
5	25	0	GAMSPATH	dH:-682.830(6SF)kJ	2	CRV	12638

Reaction No. 54971

3<Fe+3_OH-1(2)> + H+1_AsO4-3 + 5<H+1> + 2<e-1> = 2<Fe+2> + Fe+3_AsO4-3 + 6<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.(2SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-39.2(0.1SD)	0	ENS	5480

Reaction No. 54972

As(s) + 1.5<H2(g)> - 6<e-1> = As+3_H+1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:23.8(0.1SD)	4	ENS	5020

Reaction No. 55179

As+3(2)_S-2(3) + 2<OH-1> = As+3_S-2(2) + As+3_OH-1(2)_S-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0:0.05	Inf. Dilution	lgK:2.2(0.1SD)	0	ENS	140

Reaction No. 55582

Zn+2(5)_H+1(2)_AsO4-3(4)_H2O(6)_(s) + 6<H+1> = 5<Zn+2> + 4<H+1(2)_AsO4-3> + 6<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.9(0.1SD)	4	ENS	5522

Reaction No. 55583

Zn+2(3)_AsO4-3(2)_H2O(8)_(s) + 4<H+1> = 3<Zn+2> + 2<H+1(2)_AsO4-3> + 8<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.9(0.1SD)	4	ENS	5522

Reaction No. 55584

Cu+2 + H+1(3)_AsO4-3 + H2O = Cu+2_H+1_AsO4-3_H2O_(s) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.0(0.1SD)	4	ENS	5522

Reaction No. 55585

Zn+2(2)_OH-1_AsO4-3_(s) + 2<H+1> = 2<Zn+2> + H+1_AsO4-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.06(0.1SD)	4	ENS	5522

Reaction No. 55586

Cu+2(5)_H+1(2)_AsO4-3(4)_H2O(4)_(s) + 6<H+1> = 5<Cu+2> + 4<H+1(2)_AsO4-3> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(0.1SD)	4	ENS	5522

Reaction No. 55587

Cu+2(2)_OH-1_AsO4-3_(s) + 3<H+1> = 2<Cu+2> + H+1(2)_AsO4-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.0(0.1SD)	4	ENS	5522

Reaction No. 55588

Ni+2_H+1_AsO4-3_H2O(2)_(s) + H+1 = Ni+2 + H+1(2)_AsO4-3 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.8(0.1SD)	4	ENS	5522

Reaction No. 55589

Ni+2(3)_AsO4-3(2)_H2O(8)_(s) + 4<H+1> = 3<Ni+2> + 2<H+1(2)_AsO4-3> + 8<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:9.9(0.1SD)	0	ENS	5522

Reaction No. 55590

Mg+2_H+1_AsO4-3_H2O(7)_(s) + H+1 = Mg+2 + H+1(2)_AsO4-3 + 7<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(0.1SD)	4	ENS	5522

Reaction No. 55591

Mg+2(3)_AsO4-3(2)_H2O(10)_(s) + 2<H+1> = 3<Mg+2> + 2<H+1_AsO4-3> + 10<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.0(0.1SD)	4	ENS	5522

Reaction No. 55592

Ca+2_H+1(4)_AsO4-3(2)_(s) = Ca+2 + 2<H+1(2)_AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.1(0.1SD)	4	ENS	5522

Reaction No. 55593

Ca+2(5)_H+1(2)_AsO4-3(4)_(s) + 2<H+1> = 5<Ca+2> + 4<H+1_AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.8(0.1SD)	4	ENS	5522

Reaction No. 55594

Ca+2(3)_AsO4-3(2)_(s) + 2<H+1> = 3<Ca+2> + 2<H+1_AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.2(0.1SD)	0	ENS	5522

Reaction No. 55595

Ca+2(2)_OH-1_AsO4-3_(s) + H+1 = 2<Ca+2> + AsO4-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.9(0.1SD)	4	ENS	5522

Reaction No. 55596

Sr+2_H+1(4)_AsO4-3(2)_(s) + 2<H+1> = Sr+2 + 2<H+1(3)_AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.8(0.1SD)	4	ENS	5522

Reaction No. 55597

Sr+2_H+1_AsO4-3_H2O_(s) + H+1 = Sr+2 + H+1(2)_AsO4-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.3(0.1SD)	4	ENS	5522

Reaction No. 55598

Sr+2(3)_AsO4-3(2)_H2O(2)_(s) + 2<H+1> = 3<Sr+2> + 2<H+1_AsO4-3> + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7(0.1SD)	4	ENS	5522

Reaction No. 55599

Ba+2_H+1(4)_AsO4-3(2)_H2O_(s) + 2<H+1> = Ba+2 + 2<H+1(3)_AsO4-3> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.7(0.1SD)	4	ENS	5522

Reaction No. 55600

Ba+2_H+1_AsO4-3_H2O_(s) + H+1 = Ba+2 + H+1(2)_AsO4-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.7(0.1SD)	4	ENS	5522

Reaction No. 55601

Ba+2(3)_AsO4-3(2)_(s) + 2<H+1> = 3<Ba+2> + 2<H+1_AsO4-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.2(0.1SD)	0	ENS	5522

Note (1): Low wgt for inter-reaction consistency

Reaction No. 55769

Tl(s) + As(s) + 2<O2(g)> = Tl+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:155.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:154.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:113.0(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:88.39(4SF)	1	ENS	6422
5	293	0	Inf. Dilution	lgK:76.99(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-211.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-226.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-225.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-224.1(4SF)	0	MTD	6422
10	293	0	Inf. Dilution	dH:-223.1(4SF)	0	MTD	6422

Reaction No. 56873

As(s) + Co(s) = CoAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-49.371(5SF)kJ	2	EOD	29489
2	25	0	Inf. Dilution	dH:-9.7(0.1SD)	2	ENS	802
3	25	0	Inf. Dilution	dH:-51.045(5SF)kJ	2	EOD	29489

Reaction No. 56874

2<As(s)> + Co(s) = CoAs2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-96.650(5SF)kJ	2	EOD	29489
2	25	0	Inf. Dilution	dH:-14.7(0.1SD)	1	ENS	802
3	25	0	Inf. Dilution	dH:-83.262(5SF)kJ	1	EOD	29489

Reaction No. 56875

2<Co(s)> + As(s) = Co2As(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-9.5(0.1SD)	5	ENS	802

Reaction No. 56876

2<Co(s)> + 3<As(s)> = Co2As3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-23.3(0.1SD)	5	ENS	802

Reaction No. 56877

3<Co(s)> + 2<As(s)> = Co3As2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-19.4(0.1SD)	5	ENS	802

Reaction No. 56878

5<Co(s)> + 2<As(s)> = Co5As2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-19.0(0.1SD)	6	ENS	802

Reaction No. 56879

2<As(s)> + 3<Co(s)> + 4<O2(g)> = Co+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:292.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:290.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:209.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:161.3(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:129.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-387.4(0.1SD)	0	ENS	802
7	25	0	Inf. Dilution	dG:-399.6(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-772.87(5SF)kJ	0	EOD	29489
9	25	0	Inf. Dilution	dH:-445.6(4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-891.19(5SF)kJ	0	EOD	29489
11	127	0	Inf. Dilution	dH:-444.8(4SF)	1	MTD	6422
12	227	0	Inf. Dilution	dH:-443.8(4SF)	0	MTD	6422
13	327	0	Inf. Dilution	dH:-442.8(4SF)	0	MTD	6422

Reaction No. 56962

Co+3_H+1(2)_NH3(5) + AsO4-3 = Co+3_H+1(2)_NH3(5)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	4.8	1	NaClO4	lgK:-4.(0.5SD)	4	ENS	5398
2	25	0	Inf. Dilution	lgK:-4.(1SF)	1	EES	

Reaction No. 56963

Co+3_H+1_NH3(5) + AsO4-3 = Co+3_H+1_NH3(5)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	4.8	1	NaClO4	lgK:-8.(0.5SD)	4	ENS	5398
2	25	0	Inf. Dilution	lgK:-8.(1SF)	1	EES	

Reaction No. 57131

As(s) + 1.5<H2(g)> + 2<O2(g)> = H+1(3)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-184.0(0.1SD)	0	ENS	5020
2	25	0	Inf. Dilution	dG:-766.0(4SF)kJ	4	CRV	9015
3	25	0	Inf. Dilution	dG:-183.100(6SF)	4	CRV	9029
4	25	0	Inf. Dilution	dG:-183.1(4SF)	4	CRV	12011
5	25	0	Inf. Dilution	dG:-766.1(4SF)kJ	5	CRV	12588
6	25	0	Inf. Dilution	dG:-766.12(4.015SD)kJ	6	CRV	14923
7	25	0	Inf. Dilution	dG:-766.75(5SF)kJ	3	CRV	29482
8	25	0	Inf. Dilution	dG:-183.10(5SF)	2	EOD	29489
9	25	0	GAMSPATH	dG:-766.3(4SF)kJ	1	CRV	12638
10	25	0	Inf. Dilution	dH:-902.5(4SF)kJ	4	CRV	9015
11	25	0	Inf. Dilution	dH:-215.700(6SF)	4	CRV	9029
12	25	0	Inf. Dilution	dH:-215.7(4SF)	4	CRV	12011
13	25	0	Inf. Dilution	dH:-902.50(4.000SD)kJ	6	CRV	14923
14	25	0	Inf. Dilution	dH:-903.45(5SF)kJ	3	CRV	29482
15	25	0	Inf. Dilution	dH:-215.70(5SF)	2	EOD	29489
16	25	0	GAMSPATH	dH:-902.490(6SF)kJ	3	CRV	12638

Reaction No. 57132

As(s) + 0.5<H2(g)> + 2<O2(g)> + 2<e-1> = H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:125.1(4SF)	2	EOR	
2	25	0	Inf. Dilution	dG:-171.5(0.1SD)	0	ENS	5020
3	25	0	Inf. Dilution	dG:-170.79(5SF)	1	CRV	6488
4	25	0	Inf. Dilution	dG:-714.6(4SF)kJ	2	CRV	7421
5	25	0	Inf. Dilution	dG:-714.60(5SF)kJ	1	CRV	9015
6	25	0	Inf. Dilution	dG:-170.790(6SF)	1	CRV	9029
7	25	0	Inf. Dilution	dG:-170.79(5SF)	1	CRV	10174
8	25	0	Inf. Dilution	dG:-170.82(5SF)	4	CRV	12011
9	25	0	Inf. Dilution	dG:-714.7(4SF)kJ	1	CRV	12588
10	25	0	Inf. Dilution	dG:-714.59(4.008SD)kJ	1	CRV	14923
11	25	0	Inf. Dilution	dG:-713.73(5SF)kJ	3	CRV	29482
12	25	0	GAMSPATH	dG:-714.7(4SF)kJ	1	CRV	12638

13	50	0	Inf. Dilution	dG:-170.73(5SF)	0	CRV	10174
14	75	0	Inf. Dilution	dG:-170.59(5SF)	0	CRV	10174
15	100	0	Inf. Dilution	dG:-170.37(5SF)	0	CRV	10174
16	125	0	Inf. Dilution	dG:-170.08(5SF)	0	CRV	10174
17	150	0	Inf. Dilution	dG:-169.70(5SF)	0	CRV	10174
18	175	0	Inf. Dilution	dG:-169.22(5SF)	0	CRV	10174
19	200	0	Inf. Dilution	dG:-168.62(5SF)	0	CRV	10174
20	225	0	Inf. Dilution	dG:-167.89(5SF)	0	CRV	10174
21	250	0	Inf. Dilution	dG:-166.97(5SF)	0	CRV	10174
22	300	0	Inf. Dilution	dG:-164.30(5SF)	0	CRV	10174
23	25	0	Inf. Dilution	dH:-216.62(5SF)	0	CRV	6488
24	25	0	Inf. Dilution	dH:-906.34(5SF)kJ	0	CRV	9015
25	25	0	Inf. Dilution	dH:-216.620(6SF)	0	CRV	9029
26	25	0	Inf. Dilution	dH:-216.62(5SF)	3	CRV	12011
27	25	0	Inf. Dilution	dH:-906.34(4.000SD)kJ	0	CRV	14923
28	25	0	Inf. Dilution	dH:-908.41(5SF)kJ	5	CRV	29482
29	25	0	GAMSPATH	dH:-906.340(6SF)kJ	0	CRV	12638

Reaction No. 57133

As(s) + H2(g) + 2<O2(g)> + e-1 = H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-181.0(0.1SD)	0	ENS	5020
2	25	0	Inf. Dilution	dG:-180.01(5SF)	4	CRV	6488
3	25	0	Inf. Dilution	dG:-753.2(4SF)kJ	5	CRV	7421
4	25	0	Inf. Dilution	dG:-753.17(5SF)kJ	4	CRV	9015
5	25	0	Inf. Dilution	dG:-180.010(6SF)	4	CRV	9029
6	25	0	Inf. Dilution	dG:-180.01(5SF)	3	CRV	10174
7	25	0	Inf. Dilution	dG:-180.04(5SF)	4	CRV	12011
8	25	0	Inf. Dilution	dG:-753.3(4SF)kJ	4	CRV	12588
9	25	0	Inf. Dilution	dG:-753.20(4.015SD)kJ	6	CRV	14923
10	25	0	Inf. Dilution	dG:-753.65(5SF)kJ	3	CRV	29482
11	25	0	GAMSPATH	dG:-753.4(4SF)kJ	1	CRV	12638
12	50	0	Inf. Dilution	dG:-180.71(5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-181.41(5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-182.12(5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-182.81(5SF)	0	CRV	10174

16	150	0	Inf. Dilution	dG:-183.50(5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-184.18(5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-184.83(5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-185.44(5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-185.99(5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-186.79(5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-217.39(5SF)	4	CRV	6488
23	25	0	Inf. Dilution	dH:-909.56(5SF)kJ	4	CRV	9015
24	25	0	Inf. Dilution	dH:-217.390(6SF)	4	CRV	9029
25	25	0	Inf. Dilution	dH:-217.39(5SF)	4	CRV	12011
26	25	0	Inf. Dilution	dH:-909.56(4.000SD)kJ	6	CRV	14923
27	25	0	Inf. Dilution	dH:-911.42(5SF)kJ	3	CRV	29482
28	25	0	GAMSPATH	dH:-909.560(6SF)kJ	3	CRV	12638

Reaction No. 57134

As(s) + 2<O2(g)> + 3<e-1> = AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.5(4SF)	2	EOR	
2	25	0	Inf. Dilution	dG:-155.4(0.1SD)	0	ENS	5020
3	25	0	Inf. Dilution	dG:-648.4(4SF)kJ	2	CRV	7421
4	25	0	Inf. Dilution	dG:-648.41(5SF)kJ	1	CRV	9015
5	25	0	Inf. Dilution	dG:-154.970(6SF)	1	CRV	9029
6	25	0	Inf. Dilution	dG:-155.00(5SF)	4	CRV	12011
7	25	0	Inf. Dilution	dG:-648.5(4SF)kJ	1	CRV	12588
8	25	0	Inf. Dilution	dG:-648.36(4.008SD)kJ	0	CRV	14923
9	25	0	Inf. Dilution	dG:-646.36(5SF)kJ	3	CRV	29482
10	25	0	GAMSPATH	dG:-648.5(4SF)kJ	0	CRV	12638
11	25	0	Inf. Dilution	dH:-888.14(5SF)kJ	0	CRV	9015
12	25	0	Inf. Dilution	dH:-212.270(6SF)	0	CRV	9029
13	25	0	Inf. Dilution	dH:-212.27(5SF)	4	CRV	12011
14	25	0	Inf. Dilution	dH:-888.14(4.000SD)kJ	0	CRV	14923
15	25	0	Inf. Dilution	dH:-890.21(5SF)kJ	5	CRV	29482
16	25	0	GAMSPATH	dH:-888.140(6SF)kJ	0	CRV	12638

Reaction No. 57135

As(s) + 2<S(s)> = As+3_S-2(2) - e-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-6.6(0.05SD)	4	ENS	5020

Reaction No. 57136							
0.5<H2(g)> + As(s) + 2<S(s)> = As+3_H+1_S-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-11.6(0.1SD)	4	ENS	5020

Reaction No. 57137							
As(s) + S(s) = AsS(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-16.8(0.1SD)	3	ENS	5020

Reaction No. 57138							
2<As(s)> + 3<S(s)> = As+3(2)_S-2(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.11(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:28.93(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:21.61(4SF)	0	ENS	6422
4	177	0	Inf. Dilution	lgK:19.07(4SF)	0	ENS	6422
5	25	0	Inf. Dilution	dG:-40.2(0.05SD)	0	ENS	5020
6	25	0	Inf. Dilution	dG:-168.6(4SF)kJ	0	CRV	6330
7	25	0	Inf. Dilution	dG:-39.72(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-84.90(4SF)kJ	5	CRV	29482
9	25	0	GAMSPATH	dG:-91.10(4SF)kJ	1	CRV	12638
10	25	0	Inf. Dilution	dH:-169.0(4SF)kJ	0	CRV	6330
11	25	0	Inf. Dilution	dH:-40.0(3SF)	0	MTD	6422
12	25	0	Inf. Dilution	dH:-85.80(4SF)kJ	5	CRV	29482
13	25	0	GAMSPATH	dH:-91.700(5SF)kJ	1	CRV	12638
14	127	0	Inf. Dilution	dH:-41.7(3SF)	0	MTD	6422
15	177	0	Inf. Dilution	dH:-42.1(3SF)	0	MTD	6422

Reaction No. 57139							
As(s) + 1.5<H2(g)> = AsH3(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.11(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:-12.04(4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:-9.201(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:-7.564(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:-6.510(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:16.5(0.1SD)	4	ENS	5020
7	25	0	Inf. Dilution	dG:68.9(3SF)kJ	4	CRV	6330
8	25	0	Inf. Dilution	dG:16.52(4SF)	4	EFE	6422
9	25	0	Inf. Dilution	dG:68.93(4SF)kJ	5	EFE	7275
10	25	0	Inf. Dilution	dH:15.9(0.1SD)	4	ENS	3006
11	25	0	Inf. Dilution	dH:66.4(3SF)kJ	4	CRV	6330
12	25	0	Inf. Dilution	dH:15.88(4SF)	4	MTD	6422
13	25	0	Inf. Dilution	dH:66.44(4SF)kJ	5	EFE	7275
14	25	0	GAMSPATH	dH:66.526(5SF)kJ	3	CRV	12638
15	127	0	Inf. Dilution	dH:15.23(4SF)	1	MTD	6422
16	227	0	Inf. Dilution	dH:14.69(4SF)	0	MTD	6422
17	327	0	Inf. Dilution	dH:14.24(4SF)	0	MTD	6422

Reaction No. 57140

2<As(s)> + 1.5<O2(g)> = As2O3(Claud.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:101.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:100.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:71.83(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:54.77(4SF)	1	ENS	6422
5	312	0	Inf. Dilution	lgK:44.88(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-140.8(0.1SD)	0	ENS	5020
7	25	0	Inf. Dilution	dG:-137.8(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-576.53(5SF)kJ	5	CRV	29482
9	25	0	Inf. Dilution	dH:-156.5(4SF)	3	MTD	6422
10	25	0	Inf. Dilution	dH:-655.67(5SF)kJ	5	CRV	29482
11	127	0	Inf. Dilution	dH:-156.3(4SF)	1	MTD	6422
12	227	0	Inf. Dilution	dH:-155.9(4SF)	0	MTD	6422
13	312	0	Inf. Dilution	dH:-155.9(4SF)	0	MTD	6422

Reaction No. 57141

H+1(3)_AsO3-3 = H+1_AsO3-3 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-21.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-21.33(0.01SD)	0	ENS	766
3	25	0	Inf. Dilution/m	lgK:-21.39(4SF)	0	CRV	24953
4	50	0	Inf. Dilution/m	lgK:-20.67(4SF)	0	CRV	24953
5	75	0	Inf. Dilution/m	lgK:-20.19(4SF)	0	CRV	24953
6	100	0	Inf. Dilution/m	lgK:-19.89(4SF)	0	CRV	24953
7	150	0	Inf. Dilution/m	lgK:-19.70(4SF)	0	CRV	24953
8	200	0	Inf. Dilution/m	lgK:-19.86(4SF)	0	CRV	24953
9	250	0	Inf. Dilution/m	lgK:-20.25(4SF)	0	CRV	24953
10	300	0	Inf. Dilution/m	lgK:-20.78(4SF)	0	CRV	24953
11	25	0	Inf. Dilution	dH:14.20(0.01SD)	0	ENS	766

Reaction No. 57142

H+1(3)_AsO3-3 = AsO3-3 + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-34.76(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-34.74(0.01SD)	4	ENS	766
3	25	0	Inf. Dilution/m	lgK:-34.78(4SF)	3	CRV	24953
4	50	0	Inf. Dilution/m	lgK:-33.75(4SF)	2	CRV	24953
5	75	0	Inf. Dilution/m	lgK:-33.06(4SF)	2	CRV	24953
6	100	0	Inf. Dilution/m	lgK:-32.63(4SF)	1	CRV	24953
7	150	0	Inf. Dilution/m	lgK:-32.34(4SF)	1	CRV	24953
8	200	0	Inf. Dilution/m	lgK:-32.56(4SF)	1	CRV	24953
9	250	0	Inf. Dilution/m	lgK:-33.09(4SF)	0	CRV	24953
10	300	0	Inf. Dilution/m	lgK:-33.83(4SF)	0	CRV	24953
11	25	0	Inf. Dilution	dH:20.25(0.01SD)	4	ENS	766

Reaction No. 57143

H+1(3)_AsO3-3 + H+1 = H+1(4)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.30(0.01SD)	3	ENS	766

Reaction No. 57144

H+1(3)_AsO4-3 = H+1_AsO4-3 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.00(0.01SD)	2	ENS	766
2	25	0	Inf. Dilution/m	lgK:-9.22(3SF)	3	CRV	24953

3	25	0.01	Unknown/m	lgK:-8.96(3SF)	2	CRV	24953
4	25	0.1	Unknown/m	lgK:-8.59(3SF)	2	CRV	24953
5	25	0.2	Unknown/m	lgK:-8.44(3SF)	2	CRV	24953
6	25	0.5	Unknown/m	lgK:-8.26(3SF)	2	CRV	24953
7	25	1	Unknown/m	lgK:-8.16(3SF)	2	CRV	24953
8	25	2	Unknown/m	lgK:-8.14(3SF)	2	CRV	24953
9	25	3	Unknown/m	lgK:-8.18(3SF)	2	CRV	24953
10	50	0	Inf. Dilution/m	lgK:-9.35(3SF)	3	CRV	24953
11	75	0	Inf. Dilution/m	lgK:-9.61(3SF)	3	CRV	24953
12	100	0	Inf. Dilution/m	lgK:-9.96(3SF)	2	CRV	24953
13	150	0	Inf. Dilution/m	lgK:-10.84(4SF)	2	CRV	24953
14	200	0	Inf. Dilution/m	lgK:-11.85(4SF)	1	CRV	24953
15	250	0	Inf. Dilution/m	lgK:-12.93(4SF)	0	CRV	24953
16	300	0	Inf. Dilution/m	lgK:-14.04(4SF)	0	CRV	24953
17	25	0	Inf. Dilution	dH:-0.92(0.01SD)	3	ENS	766

Reaction No. 57145

H+1(3)_AsO4-3 = AsO4-3 + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-20.60(0.01SD)	0	ENS	766
3	25	0	Inf. Dilution	lgK:-21.17(0.025SD)	3	EDH	29940
4	25	0	Inf. Dilution	lgK:-20.60(4SF)	3	EOD	31763
5	25	0	Inf. Dilution/m	lgK:-20.63(4SF)	0	CRV	24953
6	25	0.01	Unknown/m	lgK:-21.16(4SF)	0	CRV	24953
7	25	0.1	NaCl	lgK:-19.83(0.03SD)	3	MGL	29940
8	25	0.1	Unknown/m	lgK:-21.96(4SF)	0	CRV	24953
9	25	0.2	Unknown/m	lgK:-22.31(4SF)	0	CRV	24953
10	25	0.25	NaCl	lgK:-19.56(0.03SD)	3	MGL	29940
11	25	0.5	NaCl	lgK:-19.08(0.015SD)	4	MGL	29940
12	25	0.5	Unknown/m	lgK:-22.85(4SF)	0	CRV	24953
13	25	1	NaCl	lgK:-18.81(0.02SD)	5	MGL	29940
14	25	1	Unknown/m	lgK:-23.32(4SF)	0	CRV	24953
15	25	2	Unknown/m	lgK:-23.90(4SF)	0	CRV	24953
16	25	3	Unknown/m	lgK:-24.33(4SF)	0	CRV	24953
17	50	0	Inf. Dilution/m	lgK:-20.55(4SF)	3	CRV	24953

18	75	0	Inf. Dilution/m	lgK:-20.69(4SF)	0	CRV	24953
19	100	0	Inf. Dilution/m	lgK:-20.98(4SF)	0	CRV	24953
20	150	0	Inf. Dilution/m	lgK:-21.89(4SF)	0	CRV	24953
21	200	0	Inf. Dilution/m	lgK:-23.06(4SF)	0	CRV	24953
22	250	0	Inf. Dilution/m	lgK:-24.39(4SF)	0	CRV	24953
23	300	0	Inf. Dilution/m	lgK:-25.78(4SF)	0	CRV	24953
24	25	0	Inf. Dilution	dH:3.43(0.01SD)	0	ENS	766
25	25	0.1	NaCl	dH:29.(0.5SD)kJ	3	MCL	29940
26	25	0.7	NaCl	dH:21.(1.SD)kJ	3	MCL	29940

Reaction No. 57146

H+1(3)_AsO3-3 + 3<I-1> + 3<H+1> - 3<H2O> = As+3_I-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.15(0.01SD)	4	ENS	766
2	25	0	Inf. Dilution	dH:-1.88(0.01SD)	4	ENS	766

Reaction No. 57147

As+3(2)_S-2(3)_(s) + 6<H2O> = 2<H+1(3)_AsO3-3> + 3<H+1_S-2> + 3<H+1 >

Note: Cannot find the original sources of these discrepant data

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-60.97(0.01SD)	0	ENS	766
2	25	0	Inf. Dilution/m	lgK:-45.21(4SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-41.79(4SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:-39.00(4SF)	0	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-36.69(4SF)	0	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-33.16(4SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-30.65(4SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-28.86(4SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-27.56(4SF)	0	CRV	24953
10	25	0	Inf. Dilution	dH:82.89(0.01SD)	0	ENS	766

Reaction No. 57148

2<H+1(3)_AsO4-3> - 3<H2O> = As2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.70(0.01SD)	4	ENS	766
2	25	0	Inf. Dilution/m	lgK:-6.73(3SF)	3	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-6.43(3SF)	2	CRV	24953

4	75	0	Inf. Dilution/m	lgK:-6.21(3SF)	2	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-6.03(3SF)	1	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-5.79(3SF)	1	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-5.65(3SF)	1	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-5.58(3SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-5.56(3SF)	0	CRV	24953
10	25	0	Inf. Dilution	dH:5.40(0.01SD)	4	ENS	766

Reaction No. 57149

H+1(3)_AsO4-3 + 2<e-1> + 2<H+1> = H+1(3)_AsO3-3 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:19.44(0.01SD)	0	ENS	766
3	25	0	Inf. Dilution/m	lgK:19.44(4SF)	0	CRV	24953
4	25	0	Inf. Dilution	E:0.559(3SF)	0	CRV	18118
5	25	0	Inf. Dilution	dH:-30.02(0.01SD)	0	ENS	766

Reaction No. 57293

Fe+3 + H+1(3)_AsO4-3 = Fe+3_H+1_AsO4-3 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaNO3	lgK:0.11(0.02SD)	6	MPT	4887

Reaction No. 57295

Fe+3 + H+1(3)_AsO4-3 = Fe+3_AsO4-3 + 3<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaNO3	lgK:-1.34(0.03SD)	6	MPT	4887

Reaction No. 57336

8<H+1> + 5<MoO4-2> + H+1(2)_AsO4-3(2) = H+1(10)_MoO4-2(5)_AsO4-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:60.92(0.04MD)	5	MGL	5547

Reaction No. 57337

10<H+1> + 6<MoO4-2> + H+1(2)_AsO4-3(2) = H+1(12)_MoO4-2(6)_AsO4-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:75.25(0.04MD)	5	MGL	5547

Reaction No. 57338							
11<H+1> + 6<MoO4-2> + H+1(2)_AsO4-3(2) = H+1(13)_MoO4-2(6)_AsO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:80.73(0.05MD)	5	MGL	5547

Reaction No. 57339							
12<H+1> + 6<MoO4-2> + H+1(2)_AsO4-3(2) = H+1(14)_MoO4-2(6)_AsO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:84.07(0.07MD)	5	MGL	5547

Reaction No. 57340							
14<H+1> + 9<MoO4-2> + H+1_AsO4-3 = H+1(15)_MoO4-2(9)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:100.11(0.02MD)	5	MGL	5547

Reaction No. 57341							
15<H+1> + 9<MoO4-2> + H+1_AsO4-3 = H+1(16)_MoO4-2(9)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:103.81(0.05MD)	5	MGL	5547

Reaction No. 57342							
16<H+1> + 9<MoO4-2> + H+1_AsO4-3 = H+1(17)_MoO4-2(9)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:106.14(0.05MD)	5	MGL	5547

Reaction No. 57613							
Ca+2(10)_OH-1(2)_AsO4-3(6)_(s) + 2<H+1> = 10<Ca+2> + 6<AsO4-3> + 2<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-90.1(3SF)	2	EAE	
2	35	0	Inf. Dilution	lgK:-90.4(0.05SD)	6	MSL	5330
3	40	0	Inf. Dilution	lgK:-90.6(0.05SD)	6	MSL	5330
4	45	0	Inf. Dilution	lgK:-90.7(0.05SD)	6	MSL	5330
5	50	0	Inf. Dilution	lgK:-90.9(0.05SD)	6	MSL	5330

Reaction No. 57617							
As(s) + Fe(s) + S(s) = FeAsS(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:11.9(0.1SD)	0	ENS	3006
2	25	0	Inf. Dilution	dG:-141.60(5SF)kJ	5	CRV	29482
3	25	0	GAMSPATH	dG:-109.7(4SF)kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH:10.0(0.1SD)	0	ENS	3006
5	25	0	GAMSPATH	dH:-105.440(6SF)kJ	0	CRV	12638
Note (5): Derived from Heinrich & Eadington, Econ. Geol., 1986, 81, 511							

Reaction No. 57631							
As(s) + 0.5<O2(g)> - e-1 = AsO+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.4(3SF)	1	EOR	
2	25	0	Inf. Dilution	dG:-39.2(0.1SD)	0	ENS	3006
3	25	0	Inf. Dilution	dG:-39.15(4SF)	4	CRV	12011
4	25	0	Inf. Dilution	dG:-163.8(4SF)kJ	5	CRV	12588

Reaction No. 57632							
As(s) + O2(g) + e-1 = AsO2-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-83.7(0.1SD)	3	ENS	3006
2	25	0	Inf. Dilution	dG:-350.(3SF)kJ	5	CRV	7421
3	25	0	Inf. Dilution	dG:-349.98(5SF)kJ	4	CRV	9015
4	25	0	Inf. Dilution	dG:-83.650(5SF)	4	CRV	9029
5	25	0	Inf. Dilution	dG:-83.66(4SF)	4	CRV	12011
6	25	0	Inf. Dilution	dG:-350.0(4SF)kJ	5	CRV	12588
7	25	0	Inf. Dilution	dG:-350.02(4.008SD)kJ	6	CRV	14923
8	25	0	Inf. Dilution	dH:-102.5(0.1SD)	3	ENS	3006
9	25	0	Inf. Dilution	dH:-429.03(5SF)kJ	4	CRV	9015
10	25	0	Inf. Dilution	dH:-102.540(6SF)	4	CRV	9029
11	25	0	Inf. Dilution	dH:-102.54(5SF)	4	CRV	12011
12	25	0	Inf. Dilution	dH:-429.03(4.000SD)kJ	6	CRV	14923

Reaction No. 57633							
Cu3As(s) + 3<H+1> + 3<e-1> = 3<Cu(s)> + AsH3(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.76(0.01SD)	0	ENS	5341

Reaction No. 57634

2<Cu+2> + AsO+1 + 2<H+1> + 7<e-1> = Cu2As(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.46(0.01SD)	5	ENS	5341

Reaction No. 57635

2<Cu+2> + H+1_AsO2-1 + 3<H+1> + 7<e-1> = Cu2As(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.46(0.01SD)	5	ENS	5341

Reaction No. 57636

Cu2O(s) + H+1_AsO2-1 + 5<H+1> + 5<e-1> = Cu2As(s) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.56(0.01SD)	5	ENS	5341

Reaction No. 57638

3<Cu(s)> + As(s) = Cu3As(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.159(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:2.146(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1.626(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:1.300(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:1.075(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-2.95(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2.8(0.1SD)	4	ENS	802
8	25	0	Inf. Dilution	dH:-2.8(0.1SD)	5	ENS	802
9	25	0	Inf. Dilution	dH:-2.8(2SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-2.9(2SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-3.0(2SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-3.1(2SF)	0	MTD	6422

Reaction No. 58220

Co+3_H+1(2)_NH3(5)_AsO4-3 + H+1 = Co+3_H+1(3)_NH3(5)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:-0.5(1SF)	0	EOD	5037
2	25	0	Inf. Dilution	lgK:-0.7(1SF)	1	EES	

Reaction No. 58221							
Co+3_H+1_NH3(5)_AsO4-3 + H+1 = Co+3_H+1(2)_NH3(5)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:3.30(3SF)	0	MEF	5037
2	25	0	Inf. Dilution	lgK:3.5(2SF)	1	EES	

Reaction No. 58269							
As+3_OH-1(3) + H2O = H+1 + As+3_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.2(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.29(3SF)	0	ENS	403
3	25	0.1	KNO3	dH:6.6(0.03SD)	2	MCL	403

Reaction No. 58278							
AsO4-3 + H2O = H+1_AsO4-3 + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.4(2SF)	4	ENS	5648

Reaction No. 58279							
H+1_AsO4-3 + H2O = H+1(2)_AsO4-3 + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.1(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.2(2SF)	0	ENS	5648

Reaction No. 58280							
H+1(2)_AsO4-3 + H2O = H+1(3)_AsO4-3 + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.7(3SF)	2	ENS	5648
2	25	0	Inf. Dilution/m	lgK:-11.72(4SF)	3	CRV	24953
3	50	0	Inf. Dilution/m	lgK:-10.86(4SF)	2	CRV	24953
4	75	0	Inf. Dilution/m	lgK:-10.11(4SF)	2	CRV	24953
5	100	0	Inf. Dilution/m	lgK:-9.45(3SF)	1	CRV	24953
6	150	0	Inf. Dilution/m	lgK:-8.32(3SF)	1	CRV	24953
7	200	0	Inf. Dilution/m	lgK:-7.39(3SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:-6.62(3SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:-5.96(3SF)	0	CRV	24953

Reaction No. 59983

Co+3_NH3(5)_AsO4-3 + H+1 = Co+3_H+1_NH3(5)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	NaClO4	lgK:8.05(3SF)	4	MEF	5037
2	25	0	Inf. Dilution	lgK:8.1(2SF)	1	ESC	

Reaction No. 60243

17<H+1> + 9<MoO4-2> + H+1_AsO4-3 = H+1(18)_MoO4-2(9)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:107.48(0.05MD)	5	MSP	5547

Reaction No. 60359

As+3(6)_S-2(9)_(s) + 3<H+1(2)_S-2> = 2<As+3(3)_S-2(6)> + 6<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	Unknown	lgK:-35.(2SF)	0	CRV	2556
Note (1): Rxn deprecated coz it is thermodynamically isolated							
2	25	0	Inf. Dilution	lgK:-38.6(3SF)	0	EAE	

Reaction No. 60588

As+3_OH-1(2) + OH-1 = As+3_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	Unknown	lgK:14.8(3SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:15.1(3SF)	2	EAE	

Reaction No. 60615

As+3_OH-1(3) + H+1 + Cl-1 = As+3_Cl-1_OH-1(2) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.07(3SF)	4	CRV	2556

Reaction No. 60616

As+3_OH-1(3) + 2<H+1> + 2<Cl-1> = As+3_Cl-1(2)_OH-1 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.54(3SF)	4	CRV	2556

Reaction No. 60617

As+3_OH-1(3) + 3<H+1> + 3<Cl-1> = As+3_Cl-1(3) + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-8.7(2SF)	4	CRV	2556

Reaction No. 60705

As+3_OH-1(2) + F-1 = As+3_F-1_OH-1(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	1	Unknown	lgK:4.48(3SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:4.78(3SF)	2	EAE	

Reaction No. 65897

H+1_AsO3-3(2) + H+1 = H+1(2)_AsO3-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:8.31(3SF)	0	CRV	552
Note (1): Removed from latest version							
2	25	1	Unknown	lgK:8.44(3SF)	0	CRV	552

Reaction No. 66126

As+3_OH-1(2) + H+1_EDTA-4 = As+3_H+1_OH-1(2)_EDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.2(2SF)	0	CRV	552
2	25	0	Inf. Dilution	lgK:9.9(2SF)	1	ELC	

Reaction No. 66184

As+3_OH-1(2) + H+1_CDTA-4 = As+3_H+1_OH-1(2)_CDTA-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:20.7(0.09SD)	4	MPL	6866

Reaction No. 66247

3<Zn(s)> + 2<As(s)> + 4<O2(g)> = Zn+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:331.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:335.6(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:333.3(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:240.4(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:184.8(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:147.7(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-457.8(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-510.2(4SF)	0	MTD	6422

9	127	0	Inf. Dilution	dH:-509.6(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-508.8(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-507.8(4SF)	0	MTD	6422

Reaction No. 66248

3<Zn(s)> + 2<As(s)> = Zn3As2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.98(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:21.83(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:16.01(4SF)	2	ENS	6422
4	190	0	Inf. Dilution	lgK:13.63(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-29.99(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-32.0(3SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-32.0(3SF)	1	MTD	6422
8	190	0	Inf. Dilution	dH:-32.0(3SF)	0	MTD	6422

Reaction No. 66270

Y(s) + As(s) + 2<O2(g)> = Y+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:248.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:246.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:180.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:141.2(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:114.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-338.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-362.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-361.7(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-361.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-360.6(4SF)	0	MTD	6422

Reaction No. 66367

3<Ti(s)> + 2<As(s)> + 4<O2(g)> = Ti+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:446.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:443.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:329.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:261.7(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:216.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-608.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-625.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-623.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-620.9(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-618.2(4SF)	0	MTD	6422

Reaction No. 66399

3<Sr(s)> + 2<As(s)> + 4<O2(g)> = Sr+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:538.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:542.1(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:538.5(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:394.2(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:307.6(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:250.0(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-739.6(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-792.8(4SF)	0	MTD	6422
9	127	0	Inf. Dilution	dH:-792.4(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-791.7(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-791.1(4SF)	0	MTD	6422

Reaction No. 66417

3<Sn(s)> + 2<As(s)> + 4<O2(g)> = Sn+2(3)_AsO4-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:274.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:272.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:194.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:147.9(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:116.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-374.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-426.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-426.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-425.6(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-429.7(4SF)	0	MTD	6422

Reaction No. 66430

Sc(s) + As(s) + 2<O2(g)> = Sc+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:234.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:232.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:170.2(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:132.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:108.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-319.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-342.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-342.1(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-341.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-340.9(4SF)	0	MTD	6422

Reaction No. 66449

3<Ag(s)> + As(s) + 2<O2(g)> = Ag+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:94.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:95.55(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:94.86(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:67.29(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:50.79(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:39.83(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-130.3(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-542.6(4SF)kJ	0	CRV	9015
9	25	0	Inf. Dilution	dH:-151.6(4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-151.2(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-150.7(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-150.2(4SF)	0	MTD	6422

Reaction No. 66464

Al(s) + As(s) = AlAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.18(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:20.06(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:14.99(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:11.93(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:9.885(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-27.53(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-27.8(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-27.9(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-28.0(3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-28.2(3SF)	0	MTD	6422

Reaction No. 66465

Al(s) + As(s) + 2<O2(g)> = Al+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:233.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:232.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:169.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:132.4(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:107.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-318.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-342.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-341.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-341.1(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-340.5(4SF)	0	MTD	6422

Reaction No. 66486

As(s) + 1.5<Cl2(g)> = As+3_Cl-1(3)_(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.39(4SF)	5	ENS	6422
2	27	0	Inf. Dilution	lgK:45.06(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:31.90(4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-61.92(4SF)	5	EFE	6422
5	25	0	Inf. Dilution	dH:-72.9(3SF)	5	MTD	6422
6	127	0	Inf. Dilution	dH:-71.5(3SF)	1	MTD	6422

Reaction No. 66487

As(s) + 1.5<F2(g)> = As+3_F-1(3)_(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:135.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:134.7(4SF)	0	ENS	6422
3	57	0	Inf. Dilution	lgK:121.7(4SF)	2	ENS	6422

4	25	0	Inf. Dilution	dG:-185.0(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-196.3(4SF)	4	MTD	6422
6	57	0	Inf. Dilution	dH:-195.9(4SF)	1	MTD	6422

Reaction No. 66488

As(s) + 1.5<I2(s)> = As+3_I-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.35(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.29(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:7.648(4SF)	2	ENS	6422
4	140.6	0	Inf. Dilution	lgK:7.294(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-59.4(3SF)kJ	4	CRV	6330
6	25	0	Inf. Dilution	dG:-14.12(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-58.2(3SF)kJ	4	CRV	6330
8	25	0	Inf. Dilution	dH:-13.9(3SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-19.7(3SF)	1	MTD	6422
10	140.6	0	Inf. Dilution	dH:-19.8(3SF)	1	MTD	6422

Reaction No. 66489

2<As(s)> + 1.5<O2(g)> = As2O3(Arsen.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:100.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:100.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:71.61(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:54.50(4SF)	1	ENS	6422
5	278	0	Inf. Dilution	lgK:48.18(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-137.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-576.34(5SF)kJ	5	CRV	29482
8	25	0	GAMSPATH	dG:-576.2(4SF)kJ	3	CRV	12638
9	25	0	Inf. Dilution	dH:-157.0(4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-657.27(5SF)kJ	5	CRV	29482
11	25	0	GAMSPATH	dH:-656.890(6SF)kJ	3	CRV	12638
12	127	0	Inf. Dilution	dH:-156.8(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-156.4(4SF)	0	MTD	6422
14	278	0	Inf. Dilution	dH:-156.1(4SF)	0	MTD	6422

Reaction No. 66490

2<As(s)> + 2<S(s)> = As2S2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.47(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:24.31(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:18.09(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:14.23(4SF)	1	ENS	6422
5	307	0	Inf. Dilution	lgK:12.06(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-33.38(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-34.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-35.1(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-35.8(3SF)	0	MTD	6422
10	307	0	Inf. Dilution	dH:-36.2(3SF)	0	MTD	6422

Reaction No. 66491							
4<As(s)> + 4<S(s)> = As4S4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.57(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:22.71(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:16.66(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:12.84(4SF)	4	ENS	6422
5	307.2	0	Inf. Dilution	lgK:10.66(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-30.99(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-32.2(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-34.4(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-35.7(3SF)	4	MTD	6422
10	307.2	0	Inf. Dilution	dH:-36.5(3SF)	4	MTD	6422

Reaction No. 66492							
4<As(s)> + 4<S(s)> = As4S4(a-Realgar,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.30(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:23.15(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:17.09(4SF)	1	ENS	
4	227	0	Inf. Dilution	lgK:13.19(4SF)	0	ENS	6422
5	307.2	0	Inf. Dilution	lgK:10.96(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-127.16(5SF)	0	EFE	6422

Note (6): Was -31.79; multiplied by 4 re stoichiometry							
7	25	0	Inf. Dilution	dG:-125.2(4SF)kJ	3	CRV	29482
Note (7): Multiplied by 4 re stoichiometry							
8	25	0	GAMSPATH	dG:-83.13(4SF)kJ	0	CRV	12638
9	25	0	Inf. Dilution	dH:-33.0(3SF)	0	MTD	6422
10	25	0	Inf. Dilution	dH:-127.2(4SF)kJ	3	CRV	29482
Note (10): Multiplied by 4 re stoichiometry							
11	25	0	GAMSPATH	dH:-138.1(4SF)kJ	1	CRV	12638
Note (11): Multiplied by 4 re stoichiometry							
12	127	0	Inf. Dilution	dH:-35.2(3SF)	0	MTD	6422
13	227	0	Inf. Dilution	dH:-36.5(3SF)	0	MTD	6422
14	307.2	0	Inf. Dilution	dH:-37.3(3SF)	0	MTD	6422

Reaction No. 66493

2<As(s)> + 3<Se(s)> = As+3(2)_Se-2(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.77(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:17.66(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:13.19(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:10.47(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:8.346(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-24.24(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-24.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-24.6(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-28.9(3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-29.3(3SF)	0	MTD	6422

Reaction No. 66494

2<As(s)> + 3<Te(s)> = As+3(2)_Te-2(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.934(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:6.893(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:5.261(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:4.299(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:3.670(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-9.46(3SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-9.0(2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-8.9(2SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-8.7(2SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-8.5(2SF)	0	MTD	6422

Reaction No. 66495

3<Au(s)> + As(s) + 2<O2(g)> = Au+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.16(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:51.74(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:35.06(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:25.11(4SF)	1	ENS	6422
5	296	0	Inf. Dilution	lgK:20.32(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-71.16(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-91.8(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-91.3(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-90.8(3SF)	0	MTD	6422
10	296	0	Inf. Dilution	dH:-90.2(3SF)	0	MTD	6422

Reaction No. 66506

3<Ba(s)> + 2<As(s)> + 4<O2(g)> = Ba+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:559.3(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:555.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:406.6(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:317.3(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:257.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-763.0(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-3095.10(6SF)kJ	5	CRV	29482
8	25	0	Inf. Dilution	dH:-817.8(4SF)	2	MTD	6422
9	127	0	Inf. Dilution	dH:-817.5(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-817.6(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-818.2(4SF)	0	MTD	6422

Reaction No. 66520

3<Be(s)> + 2<As(s)> + 4<O2(g)> = Be+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:442.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:439.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:320.4(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:249.0(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:201.5(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-603.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-654.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-653.7(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-652.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-651.6(4SF)	0	MTD	6422

Reaction No. 66534

Bi(s) + As(s) + 2<O2(g)> = Bi+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:121.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:121.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:86.395(5SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:65.70(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:51.85(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-166.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-190.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-189.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-189.1(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-191.2(4SF)	0	MTD	6422

Reaction No. 66541

3<Ca(s)> + 2<As(s)> + 4<O2(g)> = Ca+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:536.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:533.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:389.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:303.4(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:246.1(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-732.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-788.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-788.0(4SF)	1	MTD	6422

9	227	0	Inf. Dilution	dH:-787.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-786.5(4SF)	0	MTD	6422

Reaction No. 66601

3<Cd(s)> + 2<As(s)> = Cd3As2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.287(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:6.242(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:4.419(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:3.327(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:2.582(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-8.58(3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-10.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-10.0(3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-10.0(3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-14.5(3SF)	0	MTD	6422

Reaction No. 66602

3<Cd(s)> + 2<As(s)> + 4<O2(g)> = Cd+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:299.9(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:297.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:213.7(4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:163.3(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:129.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-409.2(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dG:-410.2(4SF)	2	CRV	12011
8	25	0	Inf. Dilution	dH:-462.3(4SF)	2	MTD	6422
9	127	0	Inf. Dilution	dH:-461.7(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-460.9(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-464.4(4SF)	0	MTD	6422

Reaction No. 66651

Cr(s) + As(s) + 2<O2(g)> = Cr+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:170.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:169.7(4SF)	0	ENS	6422

3	27	0	Inf. Dilution	lgK:168.5(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:122.3(4SF)	1	ENS	6422
5	227	0	Inf. Dilution	lgK:94.64(4SF)	1	ENS	6422
6	327	0	Inf. Dilution	lgK:76.24(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-231.4(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-253.8(4SF)	1	MTD	6422
9	127	0	Inf. Dilution	dH:-253.4(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-252.9(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-252.3(4SF)	0	MTD	6422

Reaction No. 66652

3<Cr(s)> + 2<As(s)> + 4<O2(g)> = Cr+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:355.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:352.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:256.2(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:198.4(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:160.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-484.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-530.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-529.4(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-528.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-527.2(4SF)	0	MTD	6422

Reaction No. 66674

3<Cs(s)> + As(s) + 2<O2(g)> = Cs+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:270.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:268.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:195.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:152.1(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:122.9(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-369.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-398.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-400.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-400.0(4SF)	0	MTD	6422

10	327	0	Inf. Dilution	dH:-399.7(4SF)	0	MTD	6422

Reaction No. 66687							
3<Cu(s)> + As(s) + 2<O2(g)> = Cu+1(3)_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:109.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:108.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:77.68(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:59.22(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:46.95(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-149.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-169.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-169.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-168.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-168.0(4SF)	0	MTD	6422

Reaction No. 66688							
3<Cu(s)> + 2<As(s)> + 4<O2(g)> = Cu+2(3)_AsO4-3(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:230.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:228.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:162.9(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:123.0(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:96.70(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-314.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1300.7(5SF)kJ	0	CRV	9015
8	25	0	Inf. Dilution	dH:-363.9(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-363.2(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-362.2(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-361.1(4SF)	0	MTD	6422

Reaction No. 66707							
Fe(s) + As(s) + 2<O2(g)> = Fe+3_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:135.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:134.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:96.74(4SF)	2	ENS	6422

4	200	0	Inf. Dilution	lgK:79.32(4SF)	1	ENS	6422
5	25	0	Inf. Dilution	dG:-184.6(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-206.8(4SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-206.6(4SF)	1	MTD	6422
8	200	0	Inf. Dilution	dH:-206.4(4SF)	0	MTD	6422

Reaction No. 66708

2<As(s)> + 3<Fe(s)> + 4<O2(g)> = Fe+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:309.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:307.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:222.2(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:171.3(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:137.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-422.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1765.3(5SF)kJ	5	CRV	12588
8	25	0	Inf. Dilution	dH:-467.3(4SF)	4	MTD	6422
9	27	0	Inf. Dilution	dH:-467.2(4SF)	0	MTD	6422
10	127	0	Inf. Dilution	dH:-466.5(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-465.6(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-464.7(4SF)	0	MTD	6422

Reaction No. 66737

As(s) + Ga(s) = GaAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.33(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.25(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:8.780(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:6.679(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:5.271(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-16.82(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-16.2(3SF)	4	CRV	12011
8	25	0	Inf. Dilution	dG:-17.520(5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-17.7(3SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-17.(2SF)	4	CRV	12011
11	25	0	Inf. Dilution	dH:-18.400(5SF)	3	CRV	31291

12	127	0	Inf. Dilution	dH:-19.2(3SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-19.3(3SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-19.4(3SF)	0	MTD	6422

Reaction No. 66738

Ga(s) + As(s) + 2<O2(g)> = Ga+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:158.0(4SF)	1	ENS	6422
2	27	0	Inf. Dilution	lgK:156.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:113.1(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:86.82(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:69.35(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-218.5(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dH:-239.5(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-240.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-240.0(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-239.4(4SF)	0	MTD	6422

Reaction No. 66900

0.25<As4O6(cubic,s)> + 1.5<H2O> = As+3_OH-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.68(0.01SD)	4	CRV	6478
2	25	0:1	Unknown	lgK:-0.68(2SF)	4	ENS	773
Note (2): Sign changed							
3	25	0	Inf. Dilution	dH:3.58(0.1SD)	4	CRV	6478

Reaction No. 66933

3<Li(s)> + As(s) + 2<O2(g)> = Li+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:279.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:277.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:203.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:159.0(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:129.2(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-381.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-406.9(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-406.7(4SF)	4	MTD	6422

9	227	0	Inf. Dilution	dH:-408.8(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-408.6(4SF)	4	MTD	6422

Reaction No. 66964

2<As(s)> + 3<Mg(s)> + 4<O2(g)> = Mg+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:485.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:496.1(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:492.8(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:359.6(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:279.8(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:226.6(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-676.8(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-731.3(4SF)	0	MTD	6422
9	127	0	Inf. Dilution	dH:-731.1(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-730.4(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-729.7(4SF)	0	MTD	6422

Reaction No. 66979

As(s) + Mn(s) = MnAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.46(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.40(4SF)	4	ENS	6422
3	42.7	0	Inf. Dilution	lgK:9.907(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-14.27(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-13.6(3SF)	4	MTD	6422
6	42.7	0	Inf. Dilution	dH:-13.5(3SF)	4	MTD	6422

Reaction No. 66980

2<As(s)> + 3<Mn(s)> + 4<O2(g)> = Mn+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:379.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:377.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:274.1(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:212.4(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:171.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-517.9(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-565.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-564.9(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-564.2(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-563.3(4SF)	0	MTD	6422

Reaction No. 67008

As(s) + Mo(s) + 2<O2(g)> = Mo+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:143.3(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:142.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:102.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:78.97(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:63.22(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-194.5(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dH:-217.7(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-217.2(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-216.6(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-216.0(4SF)	0	MTD	6422

Reaction No. 67026

3<As(s)> + Na(s) = Na3As(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.78(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:32.56(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:23.53(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:17.91(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:14.15(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-44.72(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-49.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-51.2(3SF)	1	ENS	6422
9	227	0	Inf. Dilution	dH:-51.5(3SF)	0	ENS	6422
10	327	0	Inf. Dilution	dH:-51.7(3SF)	0	MTD	6422

Reaction No. 67027

As(s) + 3<Na(s)> + 2<O2(g)> = Na+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:249.8(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:248.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:181.1(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:140.7(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:113.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-340.8(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-368.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-369.9(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-369.8(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-369.5(4SF)	0	MTD	6422

Reaction No. 67098

2<As(s)> + 3<Ni(s)> + 4<O2(g)> = Ni+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:290.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:288.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:208.3(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:160.1(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:128.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-396.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-441.9(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-441.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-440.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-439.7(4SF)	0	MTD	6422

Reaction No. 67132

2<As(s)> + 4<O2(g)> + 3<Pb(s)> = Pb+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:272.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:270.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:192.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:146.3(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:115.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-371.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-425.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-425.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-424.3(4SF)	0	MTD	6422

10	327	0	Inf. Dilution	dH:-423.5(4SF)	0	MTD	6422

Reaction No. 67202							
As(s) + 2<O2(g)> + 3<Rb(s)> = Rb+1(3)_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:269.2(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:196.3(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:152.6(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:123.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-369.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-398.9(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-400.6(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-400.4(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-400.1(4SF)	0	MTD	6422

Reaction No. 67213							
7<As(s)> + 3<Re(s)> = Re3As7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.53(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:15.42(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:11.27(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:8.778(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:7.116(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-21.18(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.80(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.81(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-22.81(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-22.81(4SF)	0	MTD	6422

Reaction No. 67214							
As(s) + 2<O2(g)> + Re(s) = Re+3_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:118.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:84.47(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:64.39(4SF)	1	ENS	6422

5	327	0	Inf. Dilution	lgK:51.05(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-162.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-184.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-184.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-183.4(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-182.7(4SF)	0	MTD	6422

Reaction No. 67321

2<As(s)> + 3<Hg(l)> + 4<O2(g)> = Hg+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:181.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:179.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:124.4(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:91.33(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:69.35(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-247.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-303.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-303.2(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-302.3(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-301.1(4SF)	0	MTD	6422

Reaction No. 67337

As(s) + In(s) = InAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.335(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:9.272(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:6.713(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:5.108(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:4.007(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-12.74(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-12.8(3SF)	4	CRV	12011
8	25	0	Inf. Dilution	dG:-12.726(5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-14.00(4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-14.0(3SF)	4	CRV	12011
11	25	0	Inf. Dilution	dH:-14.000(5SF)	3	CRV	31291
12	127	0	Inf. Dilution	dH:-14.11(4SF)	1	MTD	6422

13	227	0	Inf. Dilution	dH:-15.04(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-15.18(4SF)	0	MTD	6422

Reaction No. 67338

As(s) + In(s) + 2<O2(g)> = In+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:153.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:152.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:109.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:84.01(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:66.99(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-209.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-233.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-233.5(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-233.8(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-233.2(4SF)	0	MTD	6422

Reaction No. 67371

As(s) + 3<K(s)> + 2<O2(g)> = K+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:269.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:196.7(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:153.0(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:123.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-370.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-398.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-400.5(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-400.4(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-400.1(4SF)	4	MTD	6422

Reaction No. 68334

As+3_OH-1(2) + Tartaric-2 = As+3_OH-1(2)_Tartaric-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	KCl	lgK:6.62(3SF)	5	MPL	6866

Reaction No. 68365

3<As(s)> + H2(g) + 6<S(s)> + e-1 = As+3(3)_H+1(2)_S-2(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-125.60(5SF)kJ	5	CRV	29482

Reaction No. 69864

Th+4 + H+1(5)_Arsenazo1-6 = Th+4_H+1_Arsenazo1-6 + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.1(2SF)	1	ESC	
2	35	0.1	HNO3	lgR:-6.35(3SF)	3	MGL	2332

Reaction No. 69865

Th+4_H+1_Arsenazo1-6 = Th+4_Arsenazo1-6 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.7(2SF)	1	ESC	
2	35	0.1	HNO3	lgR:-6.00(3SF)	5	MGL	2332

Reaction No. 69866

Th+4_Arsenazo1-6 + OH-1 = Th+4_OH-1_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.6(2SF)	1	ESC	
2	35	0.1	HNO3	lgK:4.45(3SF)	5	MGL	2332

Reaction No. 69867

Th+4_OH-1_Arsenazo1-6 + OH-1 = Th+4_OH-1(2)_Arsenazo1-6

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(2SF)	1	ESC	
2	35	0.1	HNO3	lgK:3.45(3SF)	5	MGL	2332

Reaction No. 70056

2<Triphenylarsine> + Ag+1 = Ag+1_Triphenylarsine(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:5.37(0.03SD)	5	MIS	2872
2	25	0.1	DMSO NH4ClO4	dH:-52.4(0.23SD)kJ	5	MCL	2872

Reaction No. 70057

3<Triphenylarsine> + Ag+1 = Ag+1_Triphenylarsine(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:6.68(0.06SD)	5	MIS	2872
2	25	0.1	DMSO NH4ClO4	dH:-98.8(0.37SD)kJ	5	MCL	2872

Reaction No. 70059

Triphenylarsine + Cu+1 = Cu+1_Triphenylarsine

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:2.65(0.02SD)	5	MIS	2863
2	25	0.1	DMSO NH4ClO4	dH:-5.33(3SF)	5	MCL	2863

Reaction No. 70060

2<Triphenylarsine> + Cu+1 = Cu+1_Triphenylarsine(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:4.04(0.02SD)	5	MIS	2863
2	25	0.1	DMSO NH4ClO4	dH:-13.4(3SF)	5	MCL	2863

Reaction No. 70061

Triphenylarsine + Hg+2 = Hg+2_Triphenylarsine

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:6.77(0.02SD)	5	MIS	2863
2	25	0.1	DMSO NH4ClO4	dH:-8.13(3SF)	5	MCL	2863

Reaction No. 70062

2<Triphenylarsine> + Hg+2 = Hg+2_Triphenylarsine(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:9.0(0.03SD)	5	MIS	2863
2	25	0.1	DMSO NH4ClO4	dH:-14.7(3SF)	5	MCL	2863

Reaction No. 70288

As(s) = As(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:261.0(4SF)kJ	5	EFE	7275
2	25	0	Inf. Dilution	dH:302.5(4SF)kJ	5	EFE	7275

Reaction No. 70289

4<As(s)> = As4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:92.4(3SF)kJ	5	EFE	7275
2	25	0	Inf. Dilution	dH:143.9(4SF)kJ	5	EFE	7275

Reaction No. 70589

Ag+1_Triphenylarsine + Triphenylarsine = Ag+1_Triphenylarsine(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:1.81(3SF)	5	MIS	4714
2	25	0.1	DMSO NH4ClO4	dH:-19.4(3SF)kJ	5	MCL	4714

Reaction No. 70590

Ag+1_Triphenylarsine(2) + Triphenylarsine = Ag+1_Triphenylarsine(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO NH4ClO4	lgK:1.31(3SF)	5	MIS	4714
2	25	0.1	DMSO NH4ClO4	dH:-44.5(3SF)kJ	5	MCL	4714

Reaction No. 70713

As(s) = As(yellow,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:14.6(3SF)kJ	4	CRV	6330

Reaction No. 70714

As(s) + 1.5<Br2(l)> = As+3_Br-1(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-197.5(4SF)kJ	4	CRV	6330

Reaction No. 71191

As2O3(Arsen.,s) + 6<H+1> + 6<e-1> = 2<As(s)> + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.234(3SF)	5	CRV	6330

Reaction No. 71563

H+1(-1)_Cacodylic-1 + H+1 = Cacodylic-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.57(3SF)	5	CRV	6330

Reaction No. 71753

2<As(s)> + 7<O2(g)> + 3<U(s)> = UO2+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-4311.(4SF)kJ	3	CRV	12638
Note (1): Wrong stoichiometry recorded in GAMSPATH database							
2	25	0	Inf. Dilution	dH:-4689.40(2.78SD)kJ	5	MCL	22854
3	25	0	GAMSPATH	dH:-4689.4(5SF)kJ	3	CRV	12638

Reaction No. 71764

As(s) + 1.5<O2(g)> + 3<e-1> = AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-441.4(4SF)kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-421.80(5SF)kJ	0	CRV	29482
4	25	0	GAMSPATH	dG:-441.4(4SF)kJ	0	CRV	12638
5	25	0	GAMSPATH	dH:-657.310(6SF)kJ	1	CRV	12638

Reaction No. 72388

As(s) + 0.5<H2(g)> + O2(g) = H+1_AsO2-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-402.66(5SF)kJ	4	CRV	9015
2	25	0	Inf. Dilution	dG:-96.240(5SF)	4	CRV	9029
3	25	0	Inf. Dilution	dG:-96.25(4SF)	4	CRV	12011
4	25	0	Inf. Dilution	dG:-402.7(4SF)kJ	5	CRV	12588
5	25	0	Inf. Dilution	dG:-402.93(4.008SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	dH:-456.5(4SF)kJ	4	CRV	9015
7	25	0	Inf. Dilution	dH:-109.110(6SF)	4	CRV	9029
8	25	0	Inf. Dilution	dH:-109.1(4SF)	4	CRV	12011
9	25	0	Inf. Dilution	dH:-456.50(4.000SD)kJ	6	CRV	14923

Reaction No. 72449

AsO+1 + H2O = H+1_AsO2-1 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.3(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-0.34(0.05SD)	0	ENS	5559

Note (2): Source checked but prob. wrong sign

Reaction No. 72677

As(s) + H2(g) + Ni(s) + 2<O2(g)> - e-1 = Ni+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-808.12(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-978.16(5SF)kJ	5	CRV	29482

Reaction No. 73535

4<As(s)> + 3<O2(g)> = As4O6(cubic,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1152.5(5SF)kJ	5	CRV	12588
2	25	0	Inf. Dilution	dG:-1152.50(16.032SD)kJ	6	CRV	14923
3	25	0	Inf. Dilution	dH:-1313.9(16.000SD)kJ	6	CRV	14923

Reaction No. 73536

4<As(s)> + 3<O2(g)> = As4O6(mono,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1154.0(5SF)kJ	5	CRV	12588
2	25	0	Inf. Dilution	dG:-1154.0(16.041SD)kJ	6	CRV	14923
3	25	0	Inf. Dilution	dH:-1309.6(16.000SD)kJ	6	CRV	14923

Reaction No. 73537

6<As(s)> + 5<H2(g)> + 10<O2(g)> = As2O5*3.5H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-4248.4(24.000SD)kJ	6	CRV	14923

Reaction No. 74265

Ag+1(3)_AsO4-3_(s) + 3<e-1> = 3<Ag(s)> + AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.4010(4SF)	0	CRV	3006

Reaction No. 74451

AsO4-3 + 5<e-1> + 8<H+1> = As(s) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.648(3SF)	0	CRV	22551

Reaction No. 74452

As+3_S-2(2) + 3<e-1> = As(s) + 2<S-2>

Note: The species S-2 does not exist in aqueous solution See reaction: H+1 + S-2 = H+1_S-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.75(2SF)	0	CRV	22551

Reaction No. 74453

As(s) + 3<e-1> + 3<H2O> = AsH3(g) + 3<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-54.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.37(3SF)	0	CRV	22551

Reaction No. 74595

2<As(s)> + 2<H2(g)> + 4.5<O2(g)> + Zr(s) = Zr(HAsO4)2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2100.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2659.3(12.0MD)kJ	6	CRV	20829

Reaction No. 74613

2<As(s)> + H2(g) + 4<O2(g)> + Zr(s) = Zr(HAsO4)2(a.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1804.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2364.7(12.1MD)kJ	6	CRV	20829

Reaction No. 74614

2<As(s)> + H2(g) + 4<O2(g)> + Zr(s) = Zr(HAsO4)2(b.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1800.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-2360.9(12.0MD)kJ	6	CRV	20829

Reaction No. 74828

2<As(s)> + 5.5<O2(g)> + 2<U(s)> = UO2+2(2)_As2O7-4_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3426.00(2.06SD)	5	MCL	22854

Reaction No. 74870

3<Mg+2> + 2<As(s)> + 8<H2O> = 10<e-1> + 16<H+1> + Mg+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-85.8(3SF)	1	ELC	

Reaction No. 75463

As(s) + Fe(s) + H2(g) + 2<O2(g)> - 2<e-1> = Fe+3_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-793.7(4SF)kJ	5	CRV	12588
2	25	0	Inf. Dilution	dG:-794.75(5SF)kJ	5	CRV	29482
3	25	0	Inf. Dilution	dH:-985.48(5SF)kJ	5	CRV	29482

Reaction No. 75464

As(s) + Fe(s) + 0.5<H2(g)> + 2<O2(g)> - e-1 = Fe+3_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-788.2(4SF)kJ	5	CRV	12588
2	25	0	Inf. Dilution	dG:-787.38(5SF)kJ	5	CRV	29482
3	25	0	Inf. Dilution	dH:-971.57(5SF)kJ	5	CRV	29482

Reaction No. 75465

As(s) + Fe(s) + 2<O2(g)> = Fe+3_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:136.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-773.6(4SF)kJ	3	CRV	12588
3	25	0	Inf. Dilution	dG:-744.17(5SF)kJ	0	CRV	29482
4	25	0	Inf. Dilution	dH:-914.40(5SF)kJ	3	CRV	29482

Reaction No. 75466

As(s) + Fe(s) + 2<H2(g)> + 3<O2(g)> = Fe+3_AsO4-3_H2O(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1280.0(5SF)kJ	2	CRV	12588
2	25	0	Inf. Dilution	dG:-1287.10(6SF)kJ	3	CRV	29482

Reaction No. 75467							
As(s) + Fe(s) + 2<H2(g)> + 3<O2(g)> = Fe+3_AsO4-3_H2O(2)_(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1267.0(5SF)kJ	2	CRV	12588
2	25	0	Inf. Dilution	dG:-1271.00(6SF)kJ	3	CRV	29482

Reaction No. 75545							
As(s) + H2(g) + K(s) + 2<O2(g)> = H+1(2)_K+1_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1035.9(5SF)kJ	5	CRV	24003

Reaction No. 75566							
As(s) + 1.5<Cl2(g)> = As+3_Cl-1(3)_(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-248.9(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:-261.5(4SF)kJ	5	CRV	6330

Reaction No. 75567							
As(s) = As(yellow,g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:261.0(4SF)kJ	5	CRV	6330
2	25	0	Inf. Dilution	dH:302.5(4SF)kJ	5	CRV	6330

Reaction No. 75953							
H+1(5)_AsO3-3(2) + H+1 = H+1(6)_AsO3-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:8.31(3SF)	3	CRV	552
2	25	1	Unknown	lgK:8.44(3SF)	3	CRV	552

Reaction No. 76622							
H+1(3)_AsO4-3 + 2<OH-1> = H+1_AsO4-3 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:18.75(4SF)	3	CRV	24953
2	50	0	Inf. Dilution/m	lgK:17.18(4SF)	2	CRV	24953
3	75	0	Inf. Dilution/m	lgK:15.81(4SF)	2	CRV	24953
4	100	0	Inf. Dilution/m	lgK:14.60(4SF)	1	CRV	24953
5	150	0	Inf. Dilution/m	lgK:12.58(4SF)	1	CRV	24953

6	200	0	Inf. Dilution/m	lgK:10.94(4SF)	1	CRV	24953
7	250	0	Inf. Dilution/m	lgK:9.58(3SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:8.42(3SF)	0	CRV	24953

Reaction No. 76623

H+1(3)_AsO4-3 + 3<OH-1> = AsO4-3 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:21.33(4SF)	3	CRV	24953
2	50	0	Inf. Dilution/m	lgK:19.25(4SF)	2	CRV	24953
3	75	0	Inf. Dilution/m	lgK:17.44(4SF)	2	CRV	24953
4	100	0	Inf. Dilution/m	lgK:15.87(4SF)	1	CRV	24953
5	150	0	Inf. Dilution/m	lgK:13.24(4SF)	1	CRV	24953
6	200	0	Inf. Dilution/m	lgK:11.12(4SF)	0	CRV	24953
7	250	0	Inf. Dilution/m	lgK:9.38(3SF)	0	CRV	24953
8	300	0	Inf. Dilution/m	lgK:7.91(3SF)	0	CRV	24953

Reaction No. 76624

As+3_OH-1(3) + OH-1 = As+3_OH-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.8(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:4.69(3SF)	0	CRV	24953
3	50	0	Inf. Dilution/m	lgK:4.31(3SF)	0	CRV	24953
4	75	0	Inf. Dilution/m	lgK:3.97(3SF)	0	CRV	24953
5	100	0	Inf. Dilution/m	lgK:3.68(3SF)	0	CRV	24953
6	150	0	Inf. Dilution/m	lgK:3.19(3SF)	0	CRV	24953
7	200	0	Inf. Dilution/m	lgK:2.78(3SF)	0	CRV	24953
8	250	0	Inf. Dilution/m	lgK:2.45(3SF)	0	CRV	24953
9	300	0	Inf. Dilution/m	lgK:2.17(3SF)	0	CRV	24953

Reaction No. 76625

H+1(3)_AsO3-3 + H2O = As+3_OH-1(4) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution/m	lgK:-9.29(3SF)	3	CRV	24953
2	25	0.01	Unknown/m	lgK:-9.20(3SF)	2	CRV	24953
3	25	0.1	Unknown/m	lgK:-9.08(3SF)	2	CRV	24953
4	25	0.2	Unknown/m	lgK:-9.04(3SF)	2	CRV	24953
5	25	0.5	Unknown/m	lgK:-9.02(3SF)	2	CRV	24953

6	25	1	Unknown/m	lgK:-9.09(3SF)	2	CRV	24953
7	25	2	Unknown/m	lgK:-9.31(3SF)	2	CRV	24953
8	25	3	Unknown/m	lgK:-9.57(3SF)	2	CRV	24953
9	50	0	Inf. Dilution/m	lgK:-8.95(3SF)	2	CRV	24953
10	75	0	Inf. Dilution/m	lgK:-8.73(3SF)	2	CRV	24953
11	100	0	Inf. Dilution/m	lgK:-8.60(3SF)	2	CRV	24953
12	150	0	Inf. Dilution/m	lgK:-8.52(3SF)	1	CRV	24953
13	200	0	Inf. Dilution/m	lgK:-8.61(3SF)	1	CRV	24953
14	250	0	Inf. Dilution/m	lgK:-8.61(3SF)	0	CRV	24953
15	300	0	Inf. Dilution/m	lgK:-9.06(3SF)	0	CRV	24953

Reaction No. 76626

As4O6(cubic,s) + 6<H2O> = 4<H+1(3)_AsO3-3>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.2(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:1.58(3SF)	0	CRV	24953
3	25	1	Unknown/m	lgK:-0.68(2SF)	0	CRV	24953
4	50	0	Inf. Dilution/m	lgK:2.08(3SF)	0	CRV	24953
5	75	0	Inf. Dilution/m	lgK:2.56(3SF)	0	CRV	24953
6	100	0	Inf. Dilution/m	lgK:3.02(3SF)	0	CRV	24953
7	150	0	Inf. Dilution/m	lgK:3.89(3SF)	0	CRV	24953
8	200	0	Inf. Dilution/m	lgK:4.68(3SF)	0	CRV	24953
9	250	0	Inf. Dilution/m	lgK:5.42(3SF)	0	CRV	24953
10	300	0	Inf. Dilution/m	lgK:6.10(3SF)	0	CRV	24953

Reaction No. 76627

Fe+3_AsO4-3_H2O(2)_(s) = Fe+3 + AsO4-3 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-26.0(3SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:-20.55(4SF)	0	CRV	24953

Reaction No. 76709

4<As(s)> + 3<O2(g)> = As4O6(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1092.70(8.1SD)kJ	5	CRV	27292
2	25	0	Inf. Dilution	dH:-1196.30(8.0SD)kJ	5	CRV	27292

Reaction No. 76752

As(s) + Sn(s) = SnAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-10.013(1.6SD)kJ	5	CRV	27323
2	25	0	Inf. Dilution	dH:-9.500(1.6SD)kJ	5	CRV	27323
3	25	0	Inf. Dilution	Cp:51.200(0.13SD)J/K	5	CRV	27323

Reaction No. 76796

As(s) + Ni(s) = NiAs(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-66.583(2.3SD)kJ	5	CRV	20828
2	25	0	Inf. Dilution	dH:-70.820(2.1SD)kJ	5	CRV	20828
3	25	0	Inf. Dilution	Cp:50.800(2.SD)J/K	5	CRV	20828

Reaction No. 76797

2<As(s)> + Ni(s) = NiAs2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-90.100(4.SD)kJ	5	CRV	20828

Reaction No. 76798

2<As(s)> + 5<Ni(s)> = Ni5As2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-237.640(11.SD)kJ	5	CRV	20828
2	25	0	Inf. Dilution	dH:-244.600(10.SD)kJ	5	CRV	20828

Reaction No. 76799

8<As(s)> + 11<Ni(s)> = Ni11As8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-715.760(20.SD)kJ	5	CRV	20828
2	25	0	Inf. Dilution	dH:-743.000(18.SD)kJ	5	CRV	20828
3	25	0	Inf. Dilution	Cp:490.000(20.SD)J/K	5	CRV	20828

Reaction No. 76800

2<As(s)> + 8<H2(g)> + 3<Ni(s)> + 8<O2(g)> = Ni+2(3)_AsO4-3(2)_H2O(8)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3491.60(4.4SD)kJ	5	CRV	20828
2	25	0	Inf. Dilution	dH:-4179.00(10.SD)kJ	5	CRV	20828

Reaction No. 76801							
As(s) + 0.5<H2(g)> + Ni(s) + 2<O2(g)> = Ni+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-776.920(2.2SD)kJ	5	CRV	20828
2	25	0	Inf. Dilution	dG:-774.39(5SF)kJ	5	CRV	29482
3	25	0	Inf. Dilution	dH:-953.93(5SF)kJ	5	CRV	29482

Reaction No. 76811							
3<Ni+2> + 8<H2O> + 2<AsO4-3> = Ni+2(3)_AsO4-3(2)_H2O(8)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.1(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:28.100(0.25SD)	5	CRV	20828
3	25	0	Inf. Dilution	dG:-160.400(1.4SD)kJ	5	CRV	20828

Reaction No. 76812							
Ni+2 + H+1_AsO4-3 = Ni+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.900(0.15SD)	5	CRV	20828
2	25	0	Inf. Dilution	dG:-16.553(0.9SD)kJ	5	CRV	20828

Reaction No. 77046							
As(s) + Ca(s) + H2(g) + 2<O2(g)> - e-1 = Ca+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:231.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1314.50(6SF)kJ	0	CRV	29482
3	25	0	Inf. Dilution	dH:-1456.10(6SF)kJ	2	CRV	29482

Reaction No. 77047							
As(s) + Fe(s) + H2(g) + 2<O2(g)> - e-1 = Fe+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-860.62(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1019.70(6SF)kJ	5	CRV	29482

Reaction No. 77054							
Ga+3 + AsO4-3 = Ga+3_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:18.(2SF)	1	ELC	
2	25	5	H2SO4	dH:18.6(3SF)kJ	2	MTD	

Note (2): Souleiman et al., J. Cryst. Growth, 2014, 397, 29-38

Reaction No. 77213

As(s) + H2(g) + Na(s) + 2<O2(g)> = H+1(2)_Na+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1004.90(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1140.60(6SF)kJ	5	CRV	29482

Reaction No. 77215

As(s) + H2(g) + K(s) + 2<O2(g)> = H+1(2)_K+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1024.80(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1147.90(6SF)kJ	5	CRV	29482

Reaction No. 77216

As(s) + H2(g) + 2<O2(g)> + Sr(s) - e-1 = Sr+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:232.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1321.70(6SF)kJ	0	CRV	29482
3	25	0	Inf. Dilution	dH:-1455.40(6SF)kJ	2	CRV	29482

Reaction No. 77265

As(s) + Co(s) + H2(g) + 2<O2(g)> - e-1 = Co+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-809.14(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-972.18(5SF)kJ	5	CRV	29482

Reaction No. 77506

OH-1 + 2<Cu+2> + AsO4-3 = Cu+2(2)_OH-1_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.48(4SF)	1	ELC	

Reaction No. 77529

3<Zn+2> + 8<H2O> + 2<AsO4-3> = Zn+2(3)_AsO4-3(2)_H2O(8)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:29.06(4SF)	1	ELC	

Reaction No. 77535

Mg+2 + 7<H2O> + H+1 + AsO4-3 = Mg+2_H+1_AsO4-3_H2O(7)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.88(4SF)	1	ELC	

Reaction No. 77548

H+1 + AsO2-1 = H+1_AsO2-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.246(4SF)	1	ELC	

Reaction No. 77552

H+1 + AsO3-3 = H+1_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.4(3SF)	1	ELC	

Reaction No. 77555

2<H+1> + AsO3-3 = H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.47(4SF)	1	ELC	

Reaction No. 77559

Sr+2 + 4<H+1> + 2<AsO4-3> = Sr+2_H+1(4)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.64(4SF)	1	ELC	

Reaction No. 77560

H+1 + 2<AsO3CH3-2> = H+1_AsO3CH3-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.65(4SF)	1	ELC	

Reaction No. 77567

AsO4-3 + 2<H+1> = H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.48(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.75(0.04SD)	3	EDH	29940
3	25	0	Inf. Dilution	lgK:18.36(4SF)	3	EOD	31763

4	25	0.1	NaCl	lgK:17.70(0.025SD)	3	MGL	29940
5	25	0.25	NaCl	lgK:17.29(0.025SD)	5	MGL	29940
6	25	0.5	NaCl	lgK:17.00(0.01SD)	5	MGL	29940
7	25	1	NaCl	lgK:16.78(0.005SD)	3	MGL	29940
8	25	0.1	NaCl	dH:-24.4(0.045SD)kJ	4	MCL	29940
9	25	0.7	NaCl	dH:-23.2(0.04SD)kJ	4	MCL	29940

Reaction No. 77568

4<H+1> + AsO3-3 = H+1(4)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.46(4SF)	1	ELC	

Reaction No. 77576

H2O + H+1 + Cu+2 + AsO4-3 = Cu+2_H+1_AsO4-3_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.72(4SF)	1	ELC	

Reaction No. 77578

5<H+1> + 2<AsO3-3> = H+1(5)_AsO3-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.14(4SF)	1	ELC	

Reaction No. 77582

Ni+2 + 2<H2O> + H+1 + AsO4-3 = Ni+2_H+1_AsO4-3_H2O(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.68(4SF)	1	ELC	

Reaction No. 77583

6<H+1> + 2<AsO3-3> = H+1(6)_AsO3-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:68.81(4SF)	1	ELC	

Reaction No. 77587

H+1 + Fe+3 + AsO4-3 = Fe+3_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.33(4SF)	1	ELC	

Reaction No. 77594

3<Sr+2> + 2<H2O> + 2<AsO4-3> = Sr+2(3)_AsO4-3(2)_H2O(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.44(4SF)	1	ELC	

Reaction No. 77626

2<Zn+2> + OH-1 + AsO4-3 = Zn+2(2)_OH-1_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.51(4SF)	1	ELC	

Reaction No. 77667

Sr+2 + H2O + H+1 + AsO4-3 = Sr+2_H+1_AsO4-3_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.18(4SF)	1	ELC	

Reaction No. 77668

3<Mg+2> + 10<H2O> + 2<AsO4-3> = Mg+2(3)_AsO4-3(2)_H2O(10)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.14(4SF)	1	ELC	

Reaction No. 77699

AsO4-3 + Fe+3 = Fe+3_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:17.3(3SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							
2	25	0	Inf. Dilution	lgK:19.88(4SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:20.0(0.1SD)	3	EDH	29939
4	25	0.099	NaCl	lgK:17.68(0.02SD)	4	MSP	29939
5	25	0.122	NaCl	lgK:17.77(0.05SD)	4	MGL	29939
6	25	0.248	NaCl	lgK:17.26(0.02SD)	4	MSP	29939
7	25	0.256	NaCl	lgK:17.56(0.06SD)	2	MGL	29939
8	25	0.482	NaCl	lgK:16.68(0.01SD)	4	MSP	29939
9	25	0.486	NaCl	lgK:16.79(0.03SD)	4	MGL	29939
10	25	0.684	NaCl	lgK:16.50(0.02SD)	4	MGL	29939
11	25	0.707	NaCl	lgK:16.55(0.07SD)	4	MSP	29939
12	25	0.1	NaCl	dH:-11.(1.SD)kJ	4	MTD	29939
13	25	0.7	NaCl	dH:-15.(3.SD)kJ	4	MTD	29939

Reaction No. 77709

4<H+1> + Ca+2 + 2<AsO4-3> = Ca+2_H+1(4)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.86(4SF)	1	ELC	

Reaction No. 77721

H2O + H+1 + Ba+2 + AsO4-3 = Ba+2_H+1_AsO4-3_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.78(4SF)	1	ELC	

Reaction No. 77738

Zn+2 + H2O + H+1 + AsO4-3 = Zn+2_H+1_AsO4-3_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.22(4SF)	1	ELC	

Reaction No. 77747

2<H+1> + 5<Ca+2> + 4<AsO4-3> = Ca+2(5)_H+1(2)_AsO4-3(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.08(4SF)	1	ELC	

Reaction No. 77754

H2O + 4<H+1> + Ba+2 + 2<AsO4-3> = Ba+2_H+1(4)_AsO4-3(2)_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.74(4SF)	1	ELC	

Reaction No. 77763

OH-1 + 2<Ca+2> + AsO4-3 = Ca+2(2)_OH-1_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.1(4SF)	1	ELC	

Reaction No. 77882

2<H+1> + Ce+4 + AsO3-3 = Ce+4_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.0(4SF)	1	ELC	

Reaction No. 77886

3<H+1> + Ce+4 + AsO3-3 = Ce+4_H+1(3)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:35.64(4SF)	1	ELC	

Reaction No. 77892							
2<OH-1> + 10<Ca+2> + 6<AsO4-3> = Ca+2(10)_OH-1(2)_AsO4-3(6)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.1(4SF)	1	ELC	

Reaction No. 78184							
As(s) + H2(g) + Mn(s) + 2<O2(g)> - e-1 = Mn+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-989.44(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1133.10(6SF)kJ	5	CRV	29482

Reaction No. 78204							
As(s) + H2(g) + Mg(s) + 2<O2(g)> - e-1 = Mg+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1217.20(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1392.10(6SF)kJ	5	CRV	29482

Reaction No. 78309							
2<OH-1> + AsO+1 = AsO+1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.99(4SF)	1	ELC	

Reaction No. 78336							
As(s) + H2(g) + 0.5<O2(g)> + 2<S(s)> + e-1 = As+3_H+1_OH-1_S-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-244.40(5SF)kJ	2	CRV	29482

Reaction No. 78625							
3<Cu+1> + AsO4-3 = Cu+1(3)_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.2(4SF)	1	ELC	

Reaction No. 78912							
Y+3 + AsO4-3 = Y+3_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.73(4SF)	1	ELC	

Reaction No. 78926

Sc+3 + AsO4-3 = Sc+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.91(4SF)	1	ELC	

Reaction No. 78948

3<Cr+2> + 2<AsO4-3> = Cr+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.14(4SF)	1	ELC	

Reaction No. 78954

3<Cs+1> + AsO4-3 = Cs+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.597(4SF)	1	ELC	

Reaction No. 78963

3<Fe+2> + 2<AsO4-3> = Fe+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.36(4SF)	1	ELC	

Reaction No. 79006

3<Rb+1> + AsO4-3 = Rb+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.405(4SF)	1	ELC	

Reaction No. 79014

3<Na+1> + AsO4-3 = Na+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.328(4SF)	1	ELC	

Reaction No. 79026

La+3 + AsO4-3 = La+3_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.55(4SF)	1	ELC	

Reaction No. 79032

3<Li+1> + AsO4-3 = Li+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.99(4SF)	1	ELC	

Reaction No. 79046

3<K+1> + AsO4-3 = K+1(3)_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.4(2SF)	1	ELC	

Reaction No. 79161

3<UO2+2> + 2<AsO4-3> = UO2+2(3)_AsO4-3(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.0(3SF)	1	ELC	

Reaction No. 79359

Ni+2 + H+1 + AsO4-3 = Ni+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:12.91(4SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							
2	25	0	Inf. Dilution	lgK:14.47(4SF)	1	ELC	

Reaction No. 79413

2<H+1> + Fe+3 + AsO4-3 = Fe+3_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.92(4SF)	1	ELC	

Reaction No. 79414

2<H2O> + Fe+3 + AsO4-3 = Fe+3_AsO4-3_H2O(2)_(am.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.74(4SF)	1	ELC	

Reaction No. 79421

K+1 + 2<H+1> + AsO4-3 = H+1(2)_K+1_AsO4-3_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.48(4SF)	1	ELC	

Reaction No. 79527

Bi+3_AsO4-3_(s) = Bi+3 + AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:-24.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.35(3SF)	0	CRV	6330
Note (2): p. 8-58							

Reaction No. 79585

H+1_AsO3-3 + H+1 = H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:12.68(4SF)	3	CRV	28992
2	25	0	Inf. Dilution	lgK:12.13(4SF)	4	CRV	28992
3	50	0	Inf. Dilution	lgK:11.74(4SF)	3	CRV	28992
4	100	0	Inf. Dilution	lgK:11.30(4SF)	3	CRV	28992
5	150	0	Inf. Dilution	lgK:11.20(4SF)	3	CRV	28992
6	200	0	Inf. Dilution	lgK:11.35(4SF)	3	CRV	28992
7	250	0	Inf. Dilution	lgK:11.7(3SF)	3	CRV	28992

Reaction No. 79617

Fe+3(6)_OH-1(4)_SO4-2_AsO3-3(4)_H2O(4)_(s) + 16<H+1> = 6<Fe+3> + 4<H+1(3)_AsO3-3 > + SO4-2 + 8<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.25(1.8SD)	3	EFE	29300

Reaction No. 79618

4<As(s)> + 6<Fe(s)> + 6<H2(g)> + 12<O2(g)> + S(s) = Fe+3(6)_OH-1(4)_SO4-2_AsO3-3(4)_H2O(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5396.3(9.3SD)kJ	3	EFE	29300
2	25	0	Inf. Dilution	dH:-6196.6(8.6SD)kJ	3	EFE	29300

Reaction No. 79619

Fe+3 + H+1(2)_AsO4-3 = Fe+3_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0	Inf. Dilution	lgK:3.90(3SF)	4	EMN	29662
2	20	0	Inf. Dilution	lgK:3.99(3SF)	5	EMN	29662
3	25	0	Inf. Dilution	lgK:4.02(3SF)	5	EMN	29662
4	30	0	Inf. Dilution	lgK:4.04(3SF)	5	EMN	29662
5	40	0	Inf. Dilution	lgK:4.06(3SF)	5	EMN	29662
6	50	0	Inf. Dilution	lgK:4.05(3SF)	5	EMN	29662
7	60	0	Inf. Dilution	lgK:4.02(3SF)	5	EMN	29662

8	70	0	Inf. Dilution	lgK:3.96(3SF)	5	EMN	29662
9	80	0	Inf. Dilution	lgK:3.89(3SF)	5	EMN	29662
10	25	0	Inf. Dilution	dG:-28.5(3SF)kJ	2	EFE	29303
11	25	0	Inf. Dilution	dH:-25.53(4SF)kJ	4	EFE	29303
12	25	0	Inf. Dilution	Cp:-529.6(20.0SD)J/K	5	EMN	29662

Reaction No. 79748							
As(s) + H2(g) + 2<O2(g)> + Zn(s) - e-1 = Zn+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-903.44(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1068.50(6SF)kJ	5	CRV	29482

Reaction No. 79749							
As(s) + H2(g) + 2<O2(g)> + Pb(s) - e-1 = Pb+2_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-786.16(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-901.97(5SF)kJ	5	CRV	29482

Reaction No. 79750							
Al(s) + As(s) + H2(g) + 2<O2(g)> - 2<e-1> = Al+3_H+1(2)_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1255.10(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1467.40(6SF)kJ	5	CRV	29482

Reaction No. 79751							
As(s) + 0.5<H2(g)> + Na(s) + 2<O2(g)> + e-1 = H+1_Na+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-979.16(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1142.00(6SF)kJ	5	CRV	29482

Reaction No. 79752							
As(s) + 0.5<H2(g)> + K(s) + 2<O2(g)> + e-1 = H+1_K+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-998.95(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1151.60(6SF)kJ	5	CRV	29482

Reaction No. 79753

As(s) + 0.5<H2(g)> + Mg(s) + 2<O2(g)> = Mg+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1182.60(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1365.90(6SF)kJ	5	CRV	29482

Reaction No. 79754							
As(s) + Ca(s) + 0.5<H2(g)> + 2<O2(g)> = Ca+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:225.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1280.50(6SF)kJ	0	CRV	29482
3	25	0	Inf. Dilution	dH:-1440.70(6SF)kJ	3	CRV	29482

Reaction No. 79755							
As(s) + 0.5<H2(g)> + 2<O2(g)> + Sr(s) = Sr+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1287.60(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1443.10(6SF)kJ	5	CRV	29482

Reaction No. 79756							
As(s) + 0.5<H2(g)> + Mn(s) + 2<O2(g)> = Mn+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-960.50(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1121.50(6SF)kJ	5	CRV	29482

Reaction No. 79757							
As(s) + Fe(s) + 0.5<H2(g)> + 2<O2(g)> = Fe+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-824.08(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-995.62(5SF)kJ	5	CRV	29482

Reaction No. 79758							
As(s) + Co(s) + 0.5<H2(g)> + 2<O2(g)> = Co+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-784.57(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-959.14(5SF)kJ	5	CRV	29482

Reaction No. 79759

As(s) + Cu(s) + 0.5<H2(g)> + 2<O2(g)> = Cu+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-669.62(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-840.13(5SF)kJ	5	CRV	29482

Reaction No. 79760

As(s) + 0.5<H2(g)> + 2<O2(g)> + Zn(s) = Zn+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-877.92(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1054.90(6SF)kJ	5	CRV	29482

Reaction No. 79761

As(s) + 0.5<H2(g)> + 2<O2(g)> + Pb(s) = Pb+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-753.62(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-897.51(5SF)kJ	5	CRV	29482

Reaction No. 79762

Al(s) + As(s) + 0.5<H2(g)> + 2<O2(g)> - e-1 = Al+3_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1238.00(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1436.70(6SF)kJ	5	CRV	29482

Reaction No. 79763

As(s) + Na(s) + 2<O2(g)> + 2<e-1> = Na+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-935.96(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1062.50(6SF)kJ	5	CRV	29482

Reaction No. 79764

As(s) + K(s) + 2<O2(g)> + 2<e-1> = K+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-955.74(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1042.00(6SF)kJ	5	CRV	29482

Reaction No. 79765

As(s) + Mg(s) + 2<O2(g)> + e-1 = Mg+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1135.90(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1285.80(6SF)kJ	5	CRV	29482

Reaction No. 79766

As(s) + Ca(s) + 2<O2(g)> + e-1 = Ca+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1233.90(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1351.90(6SF)kJ	5	CRV	29482

Reaction No. 79767

As(s) + 2<O2(g)> + Sr(s) + e-1 = Sr+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:218.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1239.50(6SF)kJ	0	CRV	29482
3	25	0	Inf. Dilution	dH:-1352.40(6SF)kJ	2	CRV	29482

Reaction No. 79768

As(s) + Mn(s) + 2<O2(g)> + e-1 = Mn+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-913.32(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1036.80(6SF)kJ	5	CRV	29482

Reaction No. 79769

As(s) + Fe(s) + 2<O2(g)> + e-1 = Fe+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-781.02(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-914.38(5SF)kJ	5	CRV	29482

Reaction No. 79770

As(s) + Co(s) + 2<O2(g)> + e-1 = Co+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-741.36(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-880.65(5SF)kJ	5	CRV	29482

Reaction No. 79771

As(s) + Ni(s) + 2<O2(g)> + e-1 = Ni+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-737.66(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-884.16(5SF)kJ	5	CRV	29482

Reaction No. 79772

As(s) + Cu(s) + 2<O2(g)> + e-1 = Cu+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-634.89(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-768.21(5SF)kJ	5	CRV	29482

Reaction No. 79773

As(s) + 2<O2(g)> + Zn(s) + e-1 = Zn+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-837.31(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-978.44(5SF)kJ	5	CRV	29482

Reaction No. 79774

As(s) + 2<O2(g)> + Pb(s) + e-1 = Pb+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-710.41(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-813.52(5SF)kJ	5	CRV	29482

Reaction No. 79775

Al(s) + As(s) + 2<O2(g)> = Al+3_AsO4-3

Note: Italian supremo disagrees [DR]

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:213.6(4SF)	1	EES	
2	25	0	Inf. Dilution	dG:-1194.80(6SF)kJ	0	CRV	29482
3	25	0	Inf. Dilution	dH:-1380.80(6SF)kJ	2	CRV	29482

Reaction No. 79776

As(s) + H2(g) + Na(s) + 1.5<O2(g)> = H+1(2)_Na+1_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-850.48(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-957.43(5SF)kJ	5	CRV	29482

Reaction No. 79777

Ag(s) + As(s) + H2(g) + 1.5<O2(g)> = Ag+1_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-516.83(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-615.35(5SF)kJ	5	CRV	29482

Reaction No. 79778

As(s) + H2(g) + Mg(s) + 1.5<O2(g)> - e-1 = Mg+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1051.90(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1197.40(6SF)kJ	5	CRV	29482

Reaction No. 79779

As(s) + Ca(s) + H2(g) + 1.5<O2(g)> - e-1 = Ca+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1150.30(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1263.00(6SF)kJ	5	CRV	29482

Reaction No. 79780

As(s) + H2(g) + 1.5<O2(g)> + Sr(s) - e-1 = Sr+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1153.10(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1258.10(6SF)kJ	5	CRV	29482

Reaction No. 79781

As(s) + Ba(s) + H2(g) + 1.5<O2(g)> - e-1 = Ba+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1156.10(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1246.60(6SF)kJ	5	CRV	29482

Reaction No. 79782

As(s) + Cu(s) + H2(g) + 1.5<O2(g)> - e-1 = Cu+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-562.20(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-690.04(5SF)kJ	5	CRV	29482

Reaction No. 79783

As(s) + H2(g) + 1.5<O2(g)> + Pb(s) - e-1 = Pb+2_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-640.73(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-728.28(5SF)kJ	5	CRV	29482

Reaction No. 79784

Al(s) + As(s) + H2(g) + 1.5<O2(g)> - 2<e-1> = Al+3_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1115.50(6SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-1295.40(6SF)kJ	5	CRV	29482

Reaction No. 79785

As(s) + Fe(s) + H2(g) + 1.5<O2(g)> - 2<e-1> = Fe+3_H+1(2)_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-645.98(5SF)kJ	5	CRV	29482
2	25	0	Inf. Dilution	dH:-805.06(5SF)kJ	5	CRV	29482

Reaction No. 79786

4<As(s)> + 4<S(s)> = As4S4(b-Realgar,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-123.6(4SF)kJ	2	CRV	29482
Note (1): Multiplied by 4 re stoichiometry							
2	25	0	Inf. Dilution	dH:-124.0(4SF)kJ	2	CRV	29482
Note (2): Multiplied by 4 re stoichiometry							

Reaction No. 79787

As(s) + Ba(s) + 1.5<H2(g)> + 2.5<O2(g)> = Ba+2_H+1_AsO4-3_H2O_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1538.70(6SF)kJ	5	CRV	29482

Reaction No. 79802

As(s) + 2<S(s)> + Tl(s) = As+3_Tl+1_S-2(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-102.(3SF)kJ	4	EOD	29486

Reaction No. 79844

2<As(s)> + 3<Co(s)> + 8<H2(g)> + 8<O2(g)> = Co+2(3)_AsO4-3(2)_H2O(8)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:615.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3518.7(5SF)kJ	0	EOD	29489

Reaction No. 79962							
Fe+3 + H+1_AsO4-3 = Fe+3_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0	Inf. Dilution	lgK:10.17(4SF)	5	EMN	29662
2	20	0	Inf. Dilution	lgK:10.28(4SF)	5	EMN	29662
3	25	0	Inf. Dilution	lgK:10.35(4SF)	5	EMN	29662
4	30	0	Inf. Dilution	lgK:10.43(4SF)	5	EMN	29662
5	40	0	Inf. Dilution	lgK:10.62(4SF)	5	EMN	29662
6	50	0	Inf. Dilution	lgK:10.84(4SF)	5	EMN	29662
7	60	0	Inf. Dilution	lgK:11.09(4SF)	5	EMN	29662
8	70	0	Inf. Dilution	lgK:11.36(4SF)	4	EMN	29662
9	80	0	Inf. Dilution	lgK:11.65(4SF)	4	EMN	29662
10	25	0	Inf. Dilution	dH:25.74(4SF)kJ	5	EMN	29662
11	25	0	Inf. Dilution	Cp:843.1(54.1SD)J/K	5	EMN	29662

Reaction No. 80238							
AsO4-3 + Mg+2 = Mg+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:4.95(0.02SD)	5	MPH	29969
2	25	0	Inf. Dilution	lgK:6.41(0.03SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:5.21(0.01SD)	5	MGL	29940
4	25	0.25	NaCl	lgK:4.86(0.02SD)	5	MGL	29940
5	25	0.5	NaCl	lgK:4.66(0.03SD)	5	MGL	29940
6	25	1	NaCl	lgK:4.97(0.045SD)	5	MGL	29940
7	25	0.7	NaCl	dH:-2.(1.SD)kJ	4	MCL	29940

Reaction No. 80239							
H+1 + AsO4-3 + Mg+2 = Mg+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:12.86(4SF)	3	MPH	29969
2	25	0	Inf. Dilution	lgK:14.44(0.025SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:13.15(0.01SD)	5	MGL	29940
4	25	0.25	NaCl	lgK:12.93(0.03SD)	5	MGL	29940
5	25	0.5	NaCl	lgK:12.90(0.06SD)	4	MGL	29940

6	25	1	NaCl	lgK:13.74(0.02SD)	5	MGL	29940
7	25	0.7	NaCl	dH:-33.(1.SD)kJ	4	MCL	29940

Reaction No. 80240

2<H+1> + AsO4-3 + Mg+2 = Mg+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.40(0.01SD)	3	EDH	29940
2	25	0.1	NaCl	lgK:19.15(0.02SD)	5	MGL	29940
3	25	0.25	NaCl	lgK:18.95(0.02SD)	5	MGL	29940
4	25	0.5	NaCl	lgK:19.05(0.035SD)	5	MGL	29940
5	25	1	NaCl	lgK:19.92(0.02SD)	5	MGL	29940
6	25	0.7	NaCl	dH:-38.(1.5SD)kJ	4	MCL	29940

Reaction No. 80241

AsO4-3 + Ca+2 = Ca+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:4.33(0.03SD)	1	MPH	29969
2	25	0	Inf. Dilution	lgK:6.47(0.015SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:5.41(0.025SD)	3	MGL	29940
4	25	0.25	NaCl	lgK:5.39(0.01SD)	4	MGL	29940
5	25	0.5	NaCl	lgK:5.63(0.01SD)	5	MGL	29940
6	25	1	NaCl	lgK:6.57(0.02SD)	5	MGL	29940
7	25	0.7	NaCl	dH:-4.(1.SD)kJ	4	MCL	29940

Reaction No. 80242

H+1 + AsO4-3 + Ca+2 = Ca+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:12.54(0.02SD)	1	MPH	29969
2	25	0	Inf. Dilution	lgK:15.14(0.035SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:13.85(0.02SD)	3	MGL	29940
4	25	0.25	NaCl	lgK:13.37(0.01SD)	3	MGL	29940
5	25	0.5	NaCl	lgK:13.21(0.015SD)	4	MGL	29940
6	25	1	NaCl	lgK:13.56(0.02SD)	4	MGL	29940
7	25	0.7	NaCl	dH:-13.(1.SD)kJ	4	MCL	29940

Reaction No. 80243

2<H+1> + AsO4-3 + Ca+2 = Ca+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.70(0.005SD)	3	EDH	29940
2	25	0.1	NaCl	lgK:20.20(0.015SD)	5	MGL	29940
3	25	0.25	NaCl	lgK:19.73(0.01SD)	5	MGL	29940
4	25	0.5	NaCl	lgK:19.42(0.01SD)	5	MGL	29940
5	25	1	NaCl	lgK:19.15(0.01SD)	5	MGL	29940

Reaction No. 80244

AsO4-3 + Sr+2 = Sr+2_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:4.35(0.03SD)	1	MPH	29969
2	25	0	Inf. Dilution	lgK:7.65(0.015SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:6.31(0.01SD)	2	MGL	29940
4	25	0.25	NaCl	lgK:6.10(0.04SD)	3	MGL	29940
5	25	0.5	NaCl	lgK:6.13(0.015SD)	2	MGL	29940
6	25	1	NaCl	lgK:5.95(0.01SD)	4	MGL	29940

Reaction No. 80245

H+1 + AsO4-3 + Sr+2 = Sr+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:11.91(0.09SD)	1	MPH	29969
2	25	0	Inf. Dilution	lgK:15.32(0.015SD)	3	EDH	29940
3	25	0.1	NaCl	lgK:13.91(0.015SD)	2	MGL	29940
4	25	0.25	NaCl	lgK:13.56(0.015SD)	3	MGL	29940
5	25	0.5	NaCl	lgK:13.48(0.025SD)	3	MGL	29940
6	25	1	NaCl	lgK:13.67(0.01SD)	5	MGL	29940
7	25	0.7	NaCl	dH:-37.(1.5SD)kJ	4	MCL	29940

Reaction No. 80246

2<H+1> + AsO4-3 + Sr+2 = Sr+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.58(0.03SD)	3	EDH	29940
2	25	0.1	NaCl	lgK:20.16(0.02SD)	4	MGL	29940
3	25	0.25	NaCl	lgK:19.78(0.035SD)	5	MGL	29940
4	25	0.5	NaCl	lgK:19.46(0.025SD)	4	MGL	29940
5	25	1	NaCl	lgK:19.69(0.01SD)	5	MGL	29940
6	25	0.7	NaCl	dH:-55.(2.SD)kJ	4	MCL	29940

Reaction No. 80247							
AsO4-3 + Al+3 = Al+3_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.6(0.1SD)	3	EDH	29939
2	25	0.109	NaCl	lgK:12.68(0.03SD)	2	MGL	29939
3	25	0.25	NaCl	lgK:12.00(0.06SD)	4	MGL	29939
4	25	0.502	NaCl	lgK:11.42(4SF)	3	MGL	29939
5	25	0.697	NaCl	lgK:11.39(0.06SD)	4	MGL	29939
6	25	0.1	NaCl	dH:7.(2.SD)kJ	4	MTD	29939

Reaction No. 80248							
Al+3 + AsO4-3 + H2O = Al+3_OH-1_AsO4-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.9(0.1SD)	3	EDH	29939
2	25	0.109	NaCl	lgK:8.29(0.05SD)	4	MGL	29939
3	25	0.25	NaCl	lgK:7.87(0.08SD)	4	MGL	29939
4	25	0.502	NaCl	lgK:7.44(0.10SD)	4	MGL	29939
5	25	0.697	NaCl	lgK:7.66(0.08SD)	4	MGL	29939

Reaction No. 80250							
AsO4-3 + Ba+2 = Ba+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:3.97(3SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80251							
H+1 + AsO4-3 + Ba+2 = Ba+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:12.42(4SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80252							
AsO4-3 + Mn+2 = Mn+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:5.76(3SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80253							
H+1 + AsO4-3 + Mn+2 = Mn+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:13.25(4SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80254							
AsO4-3 + Fe+2 = Fe+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:7.902(0.009SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80255							
H+1 + AsO4-3 + Fe+2 = Fe+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:14.01(0.01SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80256							
AsO4-3 + Co+2 = Co+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:5.85(3SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80257							
H+1 + AsO4-3 + Co+2 = Co+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:13.31(4SF)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80258							
AsO4-3 + Ni+2 = Ni+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:5.272(0.008SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80259

AsO4-3 + Cu+2 = Cu+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:9.66(0.04SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80260							
AsO4-3 + Zn+2 = Zn+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:6.9(0.1SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80261							
H+1 + AsO4-3 + Zn+2 = Zn+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:13.42(0.05SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80262							
AsO4-3 + Cd+2 = Cd+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:7.43(0.04SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80263							
H+1 + AsO4-3 + Cd+2 = Cd+2_H+1_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:13.5(0.1SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80264							
AsO4-3 + Pb+2 = Pb+2_AsO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:10.8(0.2SD)	1	MPH	29969
Note (1): Data set looks systematically low cf. NaCl (should be lower but not)							

Reaction No. 80265
H+1 + AsO4-3 + Pb+2 = Pb+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:18.34(0.09SD)	1	MPH	29969

Note (1): Data set looks systematically low cf. NaCl (should be lower but not)

Reaction No. 80266

2<AsO4-3> + Fe+3 = Fe+3_AsO4-3(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	KNO3	lgK:25.3(0.2SD)	1	MPH	29969

Note (1): Data set looks systematically low cf. NaCl (should be lower but not)

Reaction No. 80605

3<Co+2> + 2<H+1_AsO2-1> + 2<H2O> = Co+2(3)_AsO3-3(2)_(s) + 6<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.544(0.16SD)	4	CRV	31301

Reaction No. 80607

3<Co+2> + 2<AsO4-3> + 8<H2O> = Co+2(3)_AsO4-3(2)_H2O(8)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.929(0.44SD)	4	CRV	31301

Reaction No. 80608

Co+2 + H+1_AsO4-3 = Co+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.874(0.17SD)	4	CRV	31301

Reaction No. 80881

UO2+2 + AsO4-3 + H+1 = UO2+2_H+1_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.76(4SF)	5	MSL	31763

Reaction No. 80882

UO2+2 + AsO4-3 + 2<H+1> = UO2+2_H+1(2)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.96(4SF)	5	MSL	31763

Reaction No. 80883

UO2+2 + AsO4-3 + 3<H+1> = UO2+2_H+1(3)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:41.53(4SF)	5	MSL	31763

Reaction No. 80884

UO2+2(3)_AsO4-3(2)_H2O(12)_(s) = 3<UO2+2> + 2<AsO4-3> + 12<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-45.32(4SF)	5	MSL	31763

Reaction No. 80965

Ph4As+1_ClO4-1_(s) = Ph4As+1 + ClO4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.40(3SF)	4	MSL	32204

Reaction No. 80966

Ph4As+1_ReO4-1_(s) = Ph4As+1 + ReO4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.49(3SF)	4	MSL	32204

Reaction No. 80967

Ph4As+1_MnO4-1_(s) = Ph4As+1 + MnO4-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.09(3SF)	4	MSL	32204

Reaction No. 80968

Ph4As+1_Ph4B-1_(s) = Ph4As+1 + Ph4B-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.30(3SF)	4	MSL	32204

Reaction No. 80969

Co+2_Ph4As+1_Cl-1(3)_(s) = Ph4As+1 + Co+2_Cl-1(3)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.17(3SF)	4	MSL	32204

Reaction No. 80970

Ph4As+1(2)_Cr2O7-2_(s) = 2<Ph4As+1> + Cr2O7-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.19(3SF)	4	MSL	32204

Reaction No. 80971

Zn+2_Ph4As+1(2)_Cl-1(4)_(s) = 2<Ph4As+1> + Zn+2_Cl-1(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.49(4SF)	4	MSL	32204

Reaction No. 81140

As(s) + 2<H2(g)> + 0.5<N2(g)> + O2(g) = H+1_NH3_AsO2-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-102.63(5SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-134.21(5SF)	4	CRV	12011

Reaction No. 81141

As(s) + 3<H2(g)> + 0.5<N2(g)> + 1.5<O2(g)> = H+1(3)_NH3_AsO3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-159.32(5SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-202.51(5SF)	4	CRV	12011

Reaction No. 81142

As(s) + 3<H2(g)> + 0.5<N2(g)> + 2<O2(g)> = H+1(3)_NH3_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-199.01(5SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-249.06(5SF)	4	CRV	12011

Reaction No. 81143

As(s) + 6<H2(g)> + 1.5<N2(g)> + 2<O2(g)> = H+1(3)_NH3(3)_AsO4-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-211.91(5SF)	4	CRV	12011
2	25	0	Inf. Dilution	dH:-307.28(5SF)	4	CRV	12011

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination

4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
EMN	Several experimental values (estim.)
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MCL	Calorimetry
MCN	Conductivity
MCO	Colorimetry (often for measuring pH)
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MGL	Glass electrode
MIS	Ion selective electrode
MKN	Rate of reaction
MMX	Mixed techniques
MOR	Optical Rotary Dispersion
MPH	pH method, not specified
MPL	Polarography
MPT	Potentiometry
MQH	Quinhydrone electrode

MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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